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Amino Acids, Peptides, and Proteins

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Amino acids and Peptides Importance

- ▶ Proteins and polypeptide monomers
- ▶ Hormones, hormone-releasing factors, neuromodulators and neurotransmitters
- ▶ the biosynthesis of porphyrins, purines, pyrimidines, and urea.
- ▶ *nutritionally essential, energy source*
- ▶ antibiotics, antitumor and toxic effects



Proteins Importance

- ▶ Enzymes, hormones, structural, transport, immunity, clotting factors, buffers



Amino acids

- ▶ more than 300 amino acids occur in nature.
- ▶ Proteins are synthesized almost exclusively from the set of 20 L- α -amino acids.



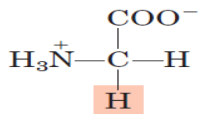
Amino acids

- ▶ All 20 of the common amino acids are α -amino acids. They have a carboxyl group and an amino group bonded to the same carbon atom (the α carbon).
- ▶ *They differ* from each other in their side chains, or R groups, which vary in structure, size, and electric charge, and which influence the solubility of the amino acids in water.

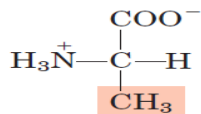


Amino acid	Abbreviation/ symbol	M_r^*	pK_a values			pI	Hydropathy index [†]	Occurrence in proteins (%) [‡]
			pK_1 (—COOH)	pK_2 (—NH ₃ ⁺)	pK_R (R group)			
Nonpolar, aliphatic R groups								
Glycine	Gly G	75	2.34	9.60		5.97	−0.4	7.2
Alanine	Ala A	89	2.34	9.69		6.01	1.8	7.8
Proline	Pro P	115	1.99	10.96		6.48	−1.6	5.2
Valine	Val V	117	2.32	9.62		5.97	4.2	6.6
Leucine	Leu L	131	2.36	9.60		5.98	3.8	9.1
Isoleucine	Ile I	131	2.36	9.68		6.02	4.5	5.3
Methionine	Met M	149	2.28	9.21		5.74	1.9	2.3
Aromatic R groups								
Phenylalanine	Phe F	165	1.83	9.13		5.48	2.8	3.9
Tyrosine	Tyr Y	181	2.20	9.11	10.07	5.66	−1.3	3.2
Tryptophan	Trp W	204	2.38	9.39		5.89	−0.9	1.4
Polar, uncharged R groups								
Serine	Ser S	105	2.21	9.15		5.68	−0.8	6.8
Threonine	Thr T	119	2.11	9.62		5.87	−0.7	5.9
Cysteine [¶]	Cys C	121	1.96	10.28	8.18	5.07	2.5	1.9
Asparagine	Asn N	132	2.02	8.80		5.41	−3.5	4.3
Glutamine	Gln Q	146	2.17	9.13		5.65	−3.5	4.2
Positively charged R groups								
Lysine	Lys K	146	2.18	8.95	10.53	9.74	−3.9	5.9
Histidine	His H	155	1.82	9.17	6.00	7.59	−3.2	2.3
Arginine	Arg R	174	2.17	9.04	12.48	10.76	−4.5	5.1
Negatively charged R groups								
Aspartate	Asp D	133	1.88	9.60	3.65	2.77	−3.5	5.3
Glutamate	Glu E	147	2.19	9.67	4.25	3.22	−3.5	6.3

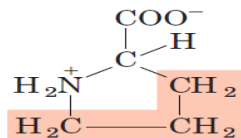
Nonpolar, aliphatic R groups



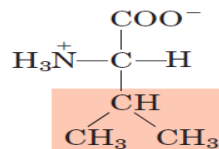
Glycine



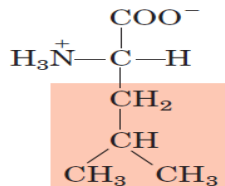
Alanine



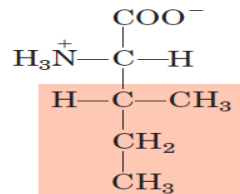
Proline



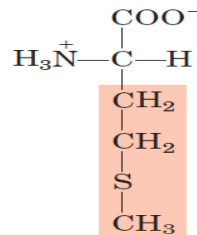
Valine



Leucine

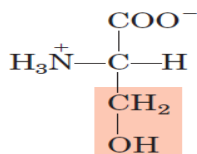


Isoleucine

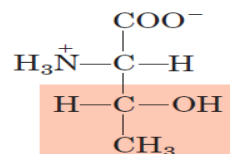


Methionine

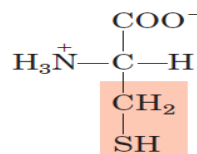
Polar, uncharged R groups



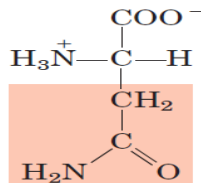
Serine



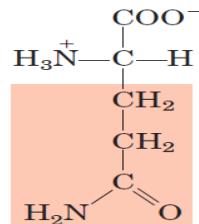
Threonine



Cysteine

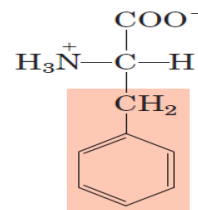


Asparagine

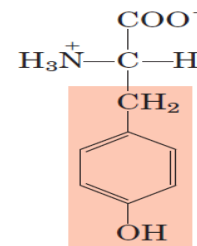


Glutamine

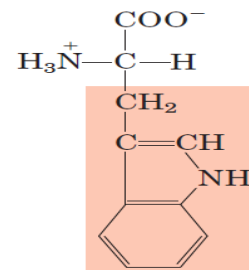
Aromatic R groups



Phenylalanine

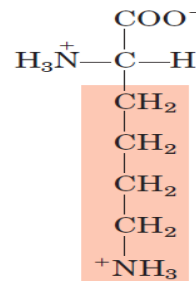


Tyrosine

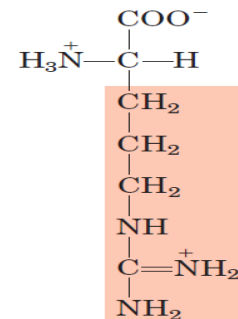


Tryptophan

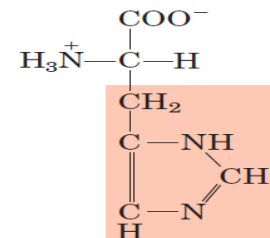
Positively charged R groups



Lysine

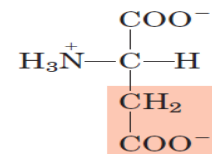


Arginine

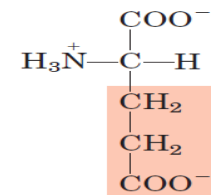


Histidine

Negatively charged R groups



Aspartate

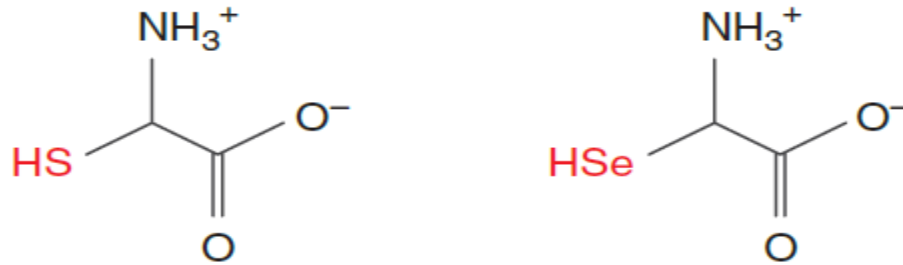


Glutamate

Hydrophilic & Hydrophobic Amino Acids

Hydrophilic	Hydrophobic
Arginine	Alanine
Asparagine	Isoleucine
Aspartic acid	Leucine
Cysteine	Methionine
Glutamic acid	Phenylalanine
Glutamine	Proline
Glycine	Tryptophan
Histidine	Tyrosine
Lysine	Valine
Serine	
Threonine	

Selenocysteine



Cysteine (left) & selenocysteine (right).

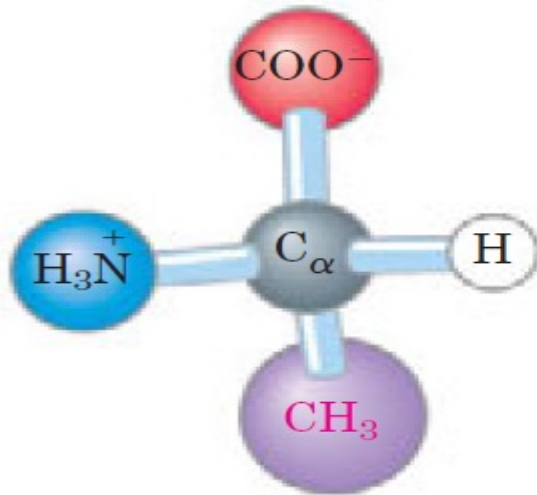


Essential amino acid

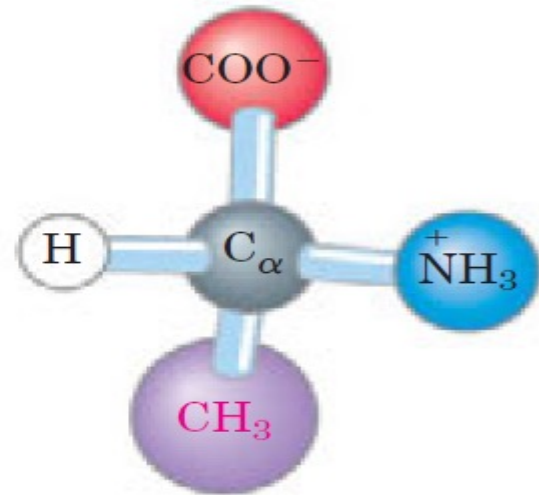
- ▶ VIP TMT HALL
- ▶ Valine Isoleucine Phenylalanine
- ▶ Threonine Methionine Tryptophan
- ▶ Histidine*, Arginine*, Leucine, Lysine



Enantiomers

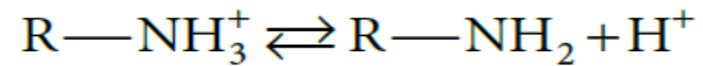
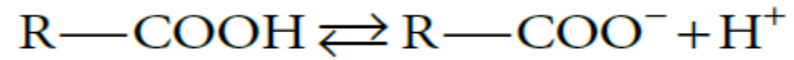


L-Alanine

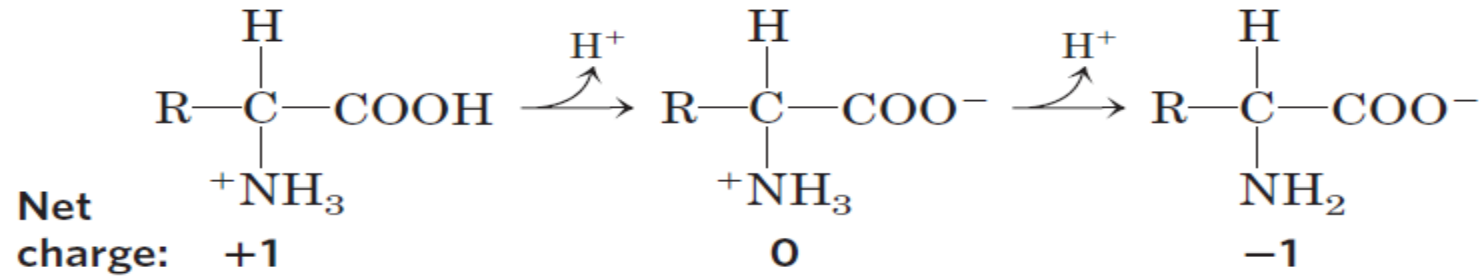


D-Alanine

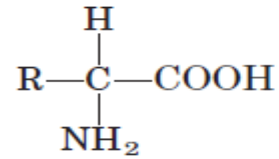
Amino acids



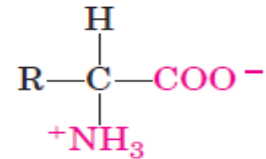
Amino acids



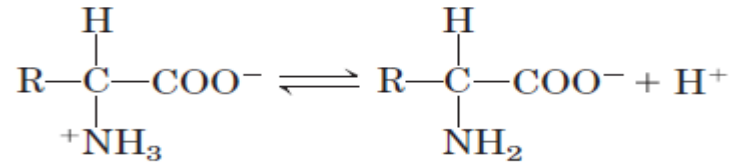
Amino acids



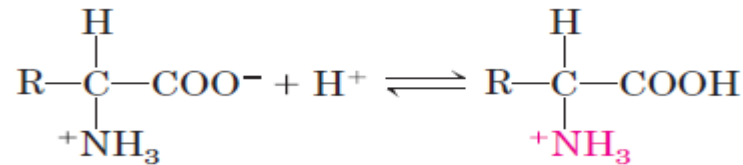
Nonionic form



Zwitterionic form



Zwitterion
as acid



Zwitterion
as base



Amino acids

- ▶ The characteristic pH at which the *net electric charge is zero is called the isoelectric point* or isoelectric pH, designated pI

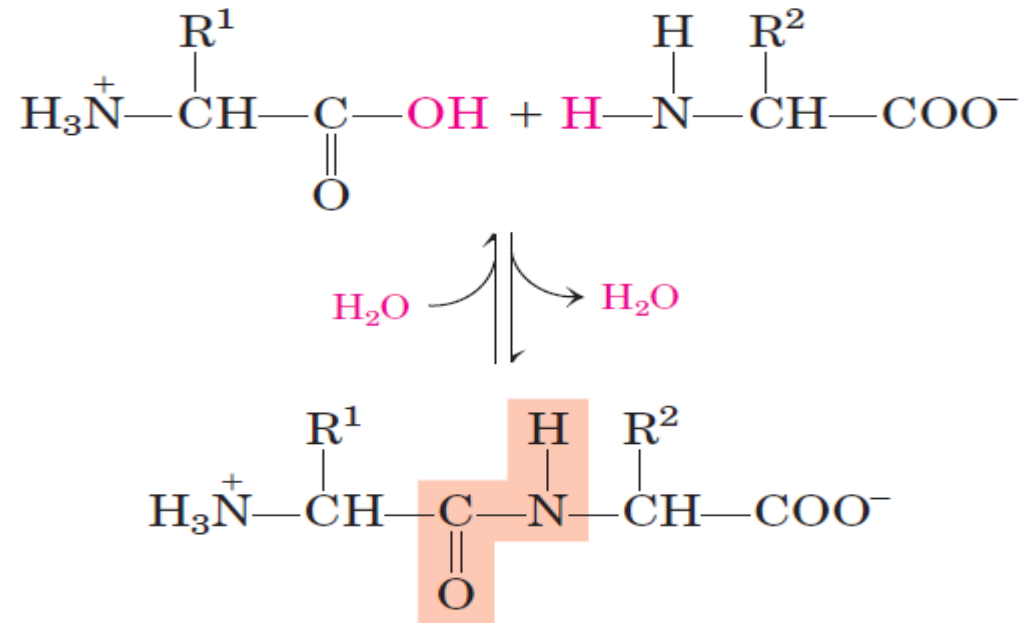


Peptides

- ▶ Amino acids are linked together by peptide bonds
- ▶ The number and order of the amino acid residues in a polypeptide constitute its primary structure.
- ▶ Peptide bonds exhibit partial double-bond character.



Peptides

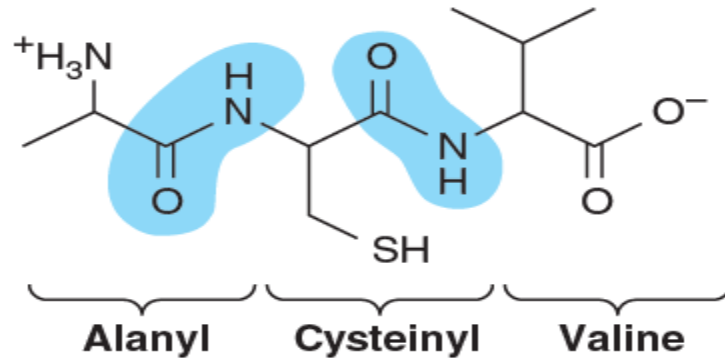


Peptides

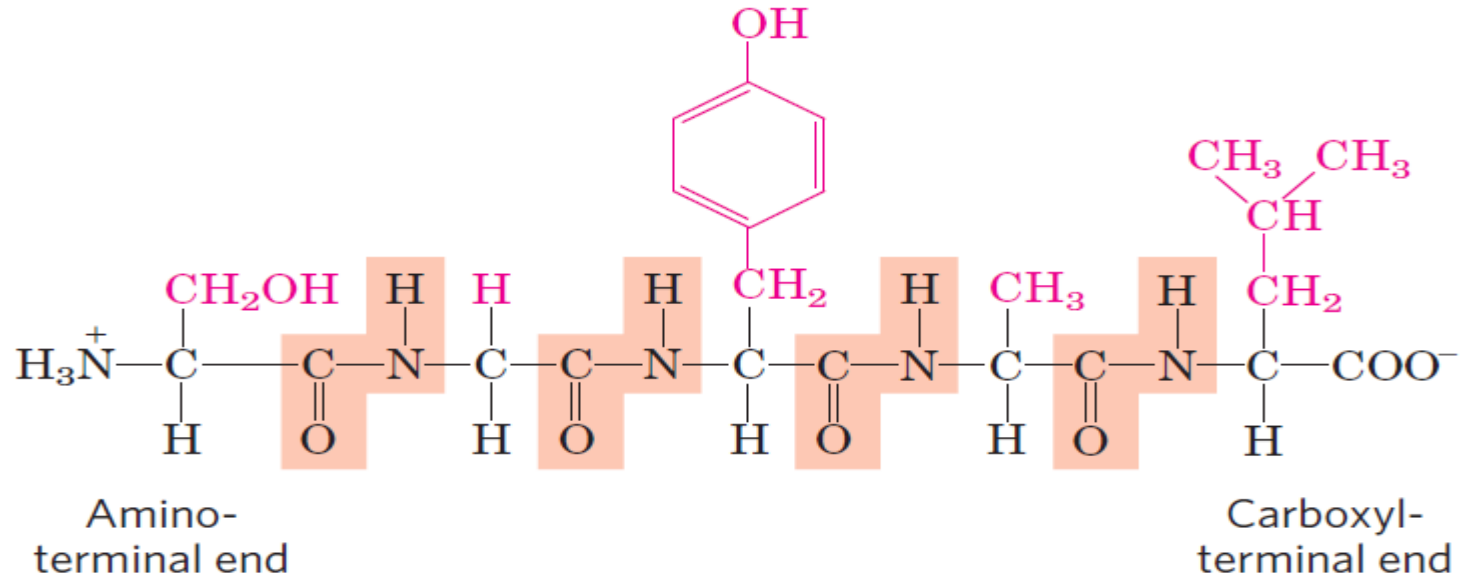
- ▶ When a few amino acids are joined in this fashion, the structure is called an oligopeptide.
- ▶ When many amino acids are joined, the product is called a polypeptide.



Peptides



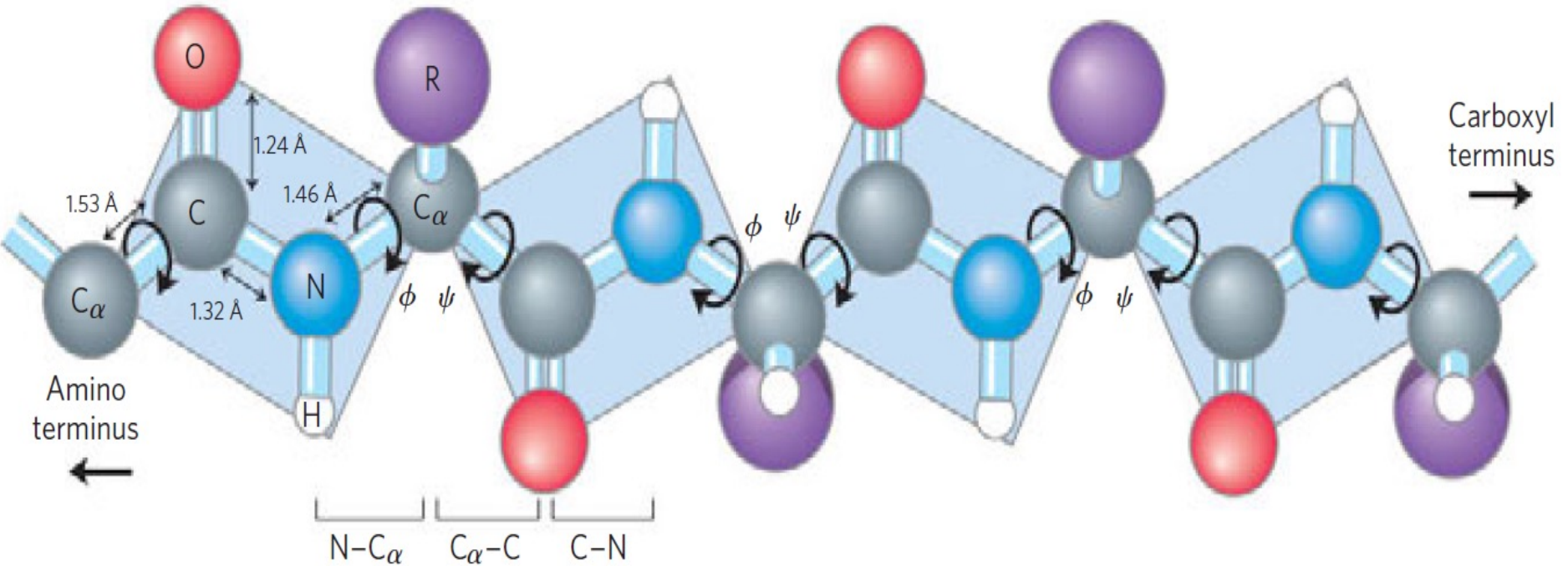
Peptides



The pentapeptide serylglycyltyrosylalanylleucine, Ser-Gly-Tyr-Ala-Leu, or SGYAL.



Peptides



Proteins

- ▶ are physically and functionally complex macromolecules that perform multiple critically important roles.

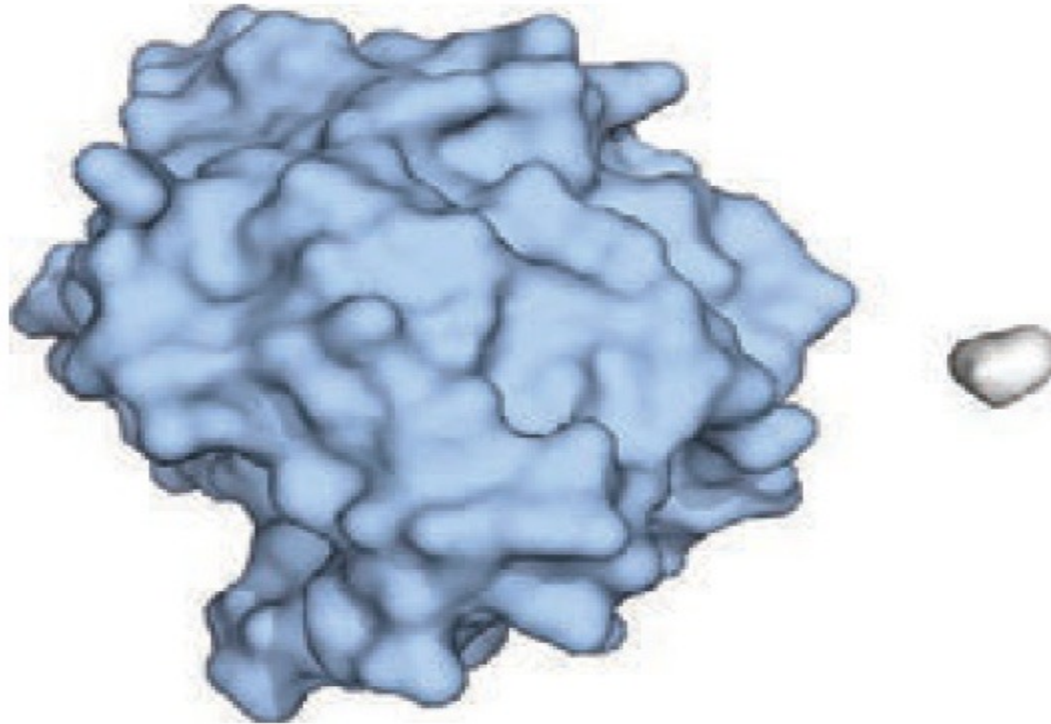


Proteins

- ▶ In nature, form follows function. In order for a newly synthesized polypeptide to mature into a biologically functional protein it must fold into a specific three-dimensional arrangement, or conformation.
- ▶ In addition, during maturation posttranslational modifications may add new chemical groups or remove transiently needed peptide segments.



Proteins



Proteins

- ▶ Each type of protein has a unique amino acid sequence that confers a particular three-dimensional structure. This structure in turn confers a unique function.
- ▶ Some proteins consist of a single polypeptide chain, but others, called multisubunit proteins, have two or more polypeptides associated noncovalently.



Proteins

- ▶ If at least two polypeptides are identical the protein is said to be oligomeric, and the identical units (consisting of one or more polypeptide chains) are referred to as protomers.



Proteins

Primary
structure

Pro
Ala
Asp
Lys
Thr
Asn
Val
Lys
Ala
Ala
Trp
Gly
Lys
Val

Secondary
structure

α Helix

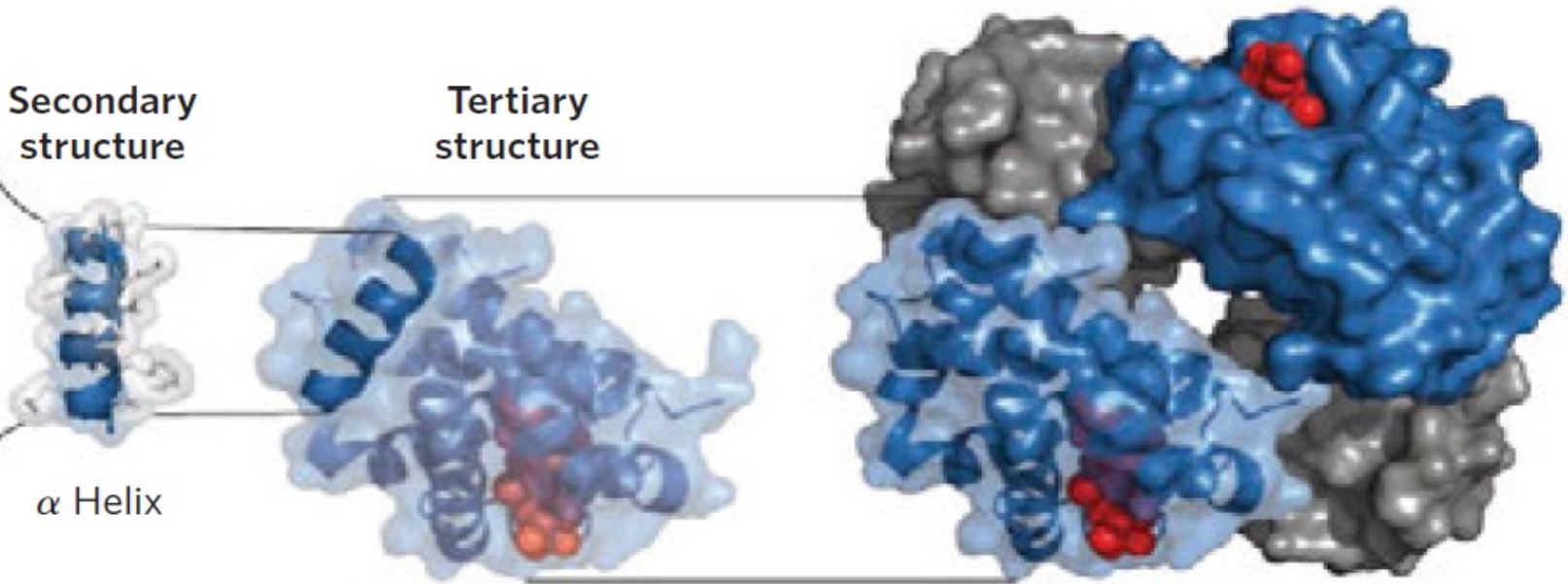
Tertiary
structure

Polypeptide chain

Quaternary
structure

Assembled subunits

Amino acid
residues



Proteins

- ▶ Although the amino acid sequence in some regions of the primary structure might vary considerably without affecting biological function, most proteins contain crucial regions that are essential to their function and whose sequence is therefore conserved.



Proteins

- ▶ Knowledge of the sequence of amino acids in a protein can offer insights into its three-dimensional structure and its function, cellular location, and evolution.



Proteins

- ▶ Some proteins contain permanently associated chemical components in addition to amino acids; these are called conjugated proteins.
- ▶ The non–amino acid part of a conjugated protein is usually called its prosthetic group.



Conjugated Proteins

Class	Prosthetic group	Example
Lipoproteins	Lipids	β_1 -Lipoprotein of blood
Glycoproteins	Carbohydrates	Immunoglobulin G
Phosphoproteins	Phosphate groups	Casein of milk
Hemoproteins	Heme (iron porphyrin)	Hemoglobin
Flavoproteins	Flavin nucleotides	Succinate dehydrogenase
Metalloproteins	Iron	Ferritin
	Zinc	Alcohol dehydrogenase
	Calcium	Calmodulin
	Molybdenum	Dinitrogenase
	Copper	Plastocyanin



ORDERS OF PROTEIN STRUCTURE

- ▶ Primary structure—the sequence of amino acids in a polypeptide chain;
- ▶ Secondary structure—the folding of short (3-30 residue), contiguous segments of polypeptide into geometrically ordered units;



ORDERS OF PROTEIN STRUCTURE

- ▶ Tertiary structure—the assembly of secondary structural units into larger functional units such as the mature polypeptide and its component domains;
- ▶ Quaternary structure—the number and types of polypeptide units of oligomeric proteins and their spatial arrangement.



Proteins

Primary
structure

Pro
Ala
Asp
Lys
Thr
Asn
Val
Lys
Ala
Ala
Trp
Gly
Lys
Val

Secondary
structure

α Helix

Tertiary
structure

Polypeptide chain

Quaternary
structure

Assembled subunits

Amino acid
residues



Secondary structure

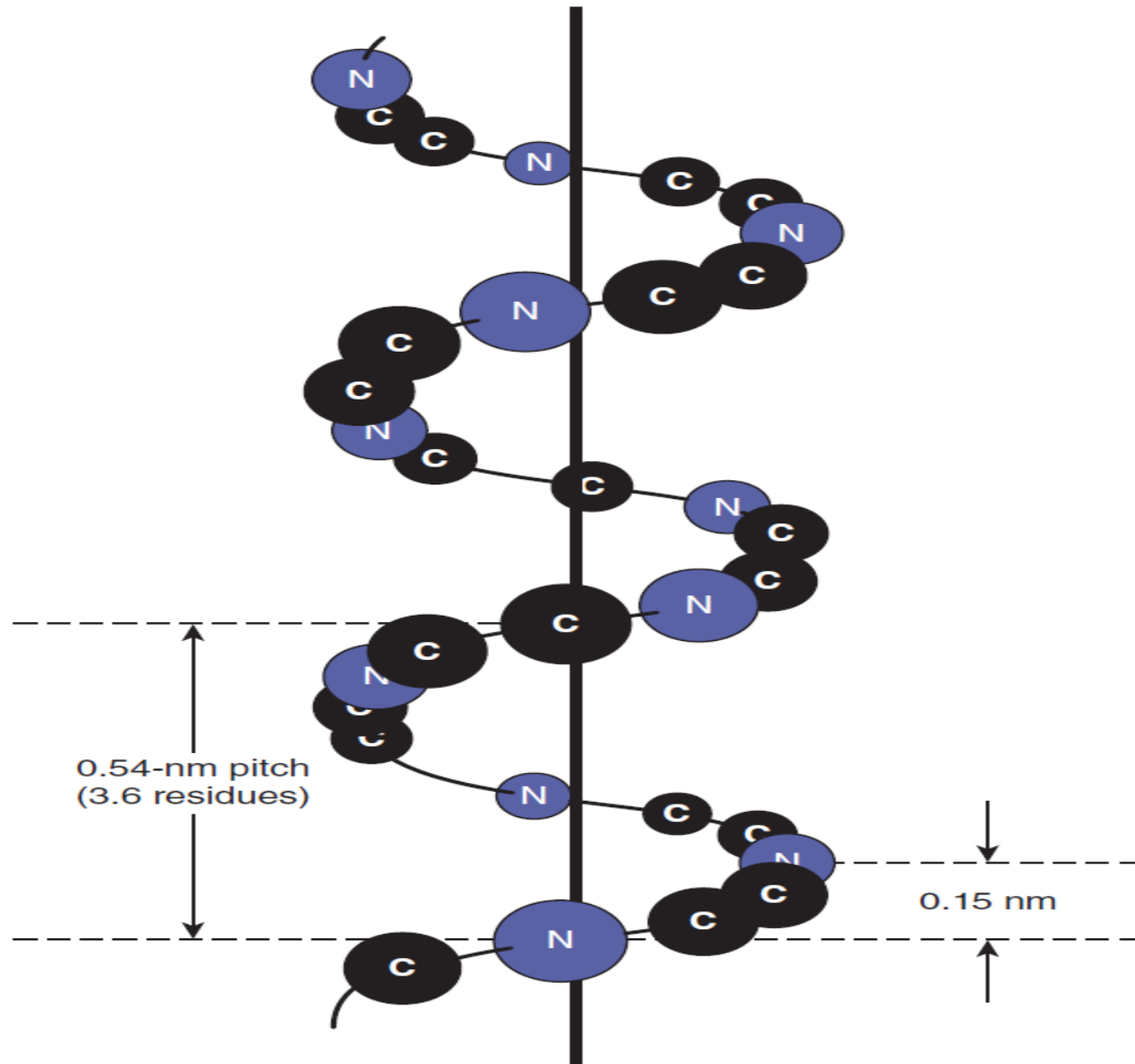
- ▶ common types of secondary structure, the **Alpha** helix and the a **Beta Sheet** sheet



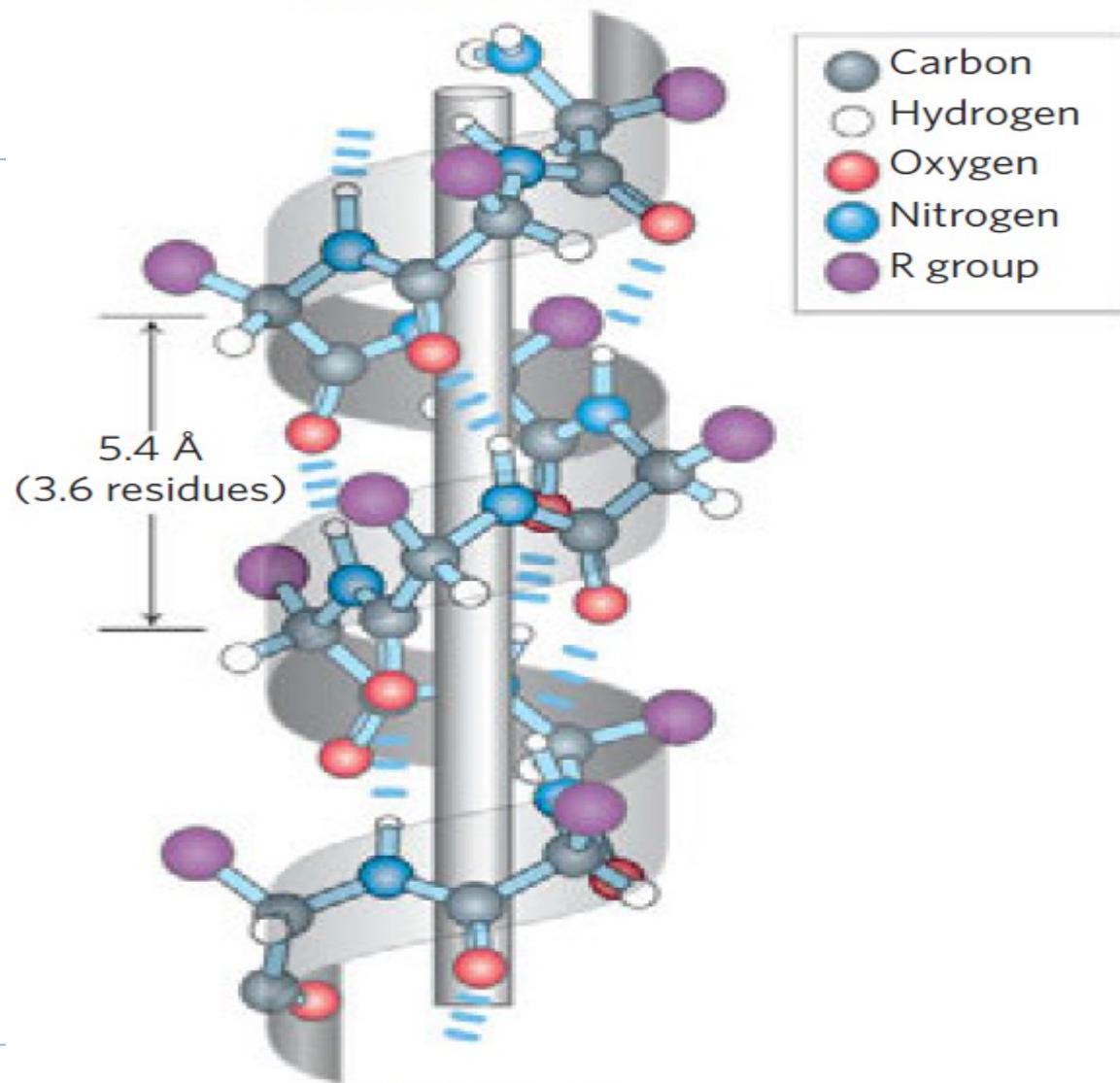
Alpha Helix

- ▶ The polypeptide backbone of an α helix is twisted by an equal amount about each α -carbon
- ▶ A complete turn of the helix contains an average of 3.6 aminoacyl residues, and the distance it rises per turn (its *pitch*) is 0.54 nm



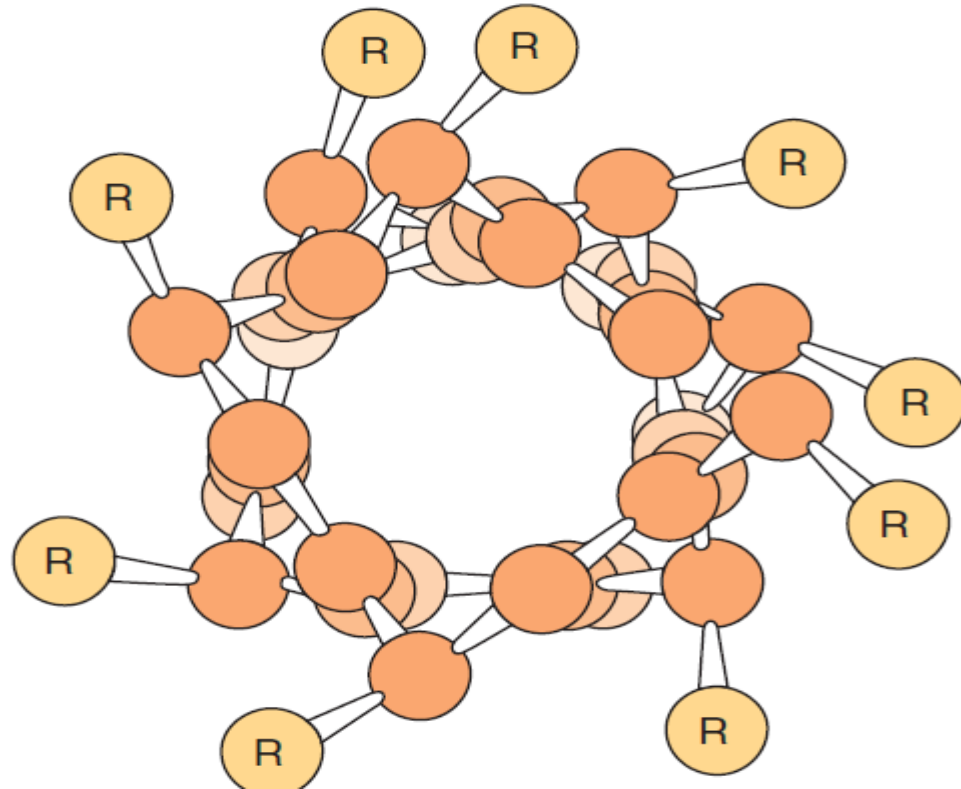


Amino terminus

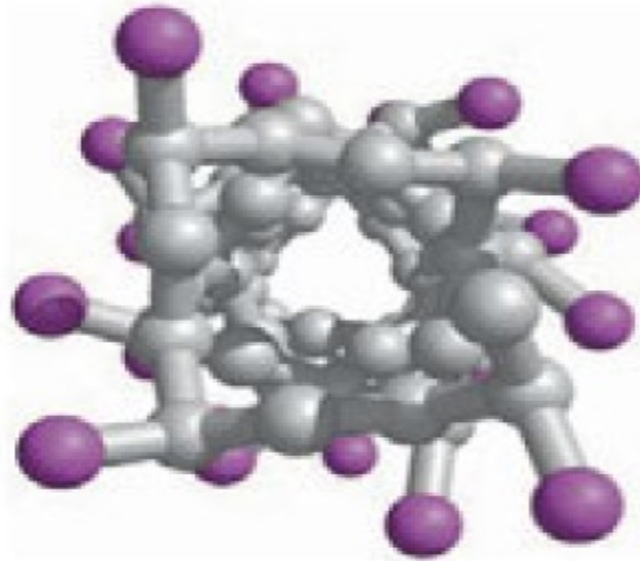


Carboxyl terminus

Alpha Helix



Alpha Helix



Alpha Helix

- ▶ Proteins contain only L-amino acids, for which a right-handed helix is by far the more stable, and only right-handed helices are present in proteins.
- ▶ Schematic diagrams of proteins represent helices as coils or cylinders

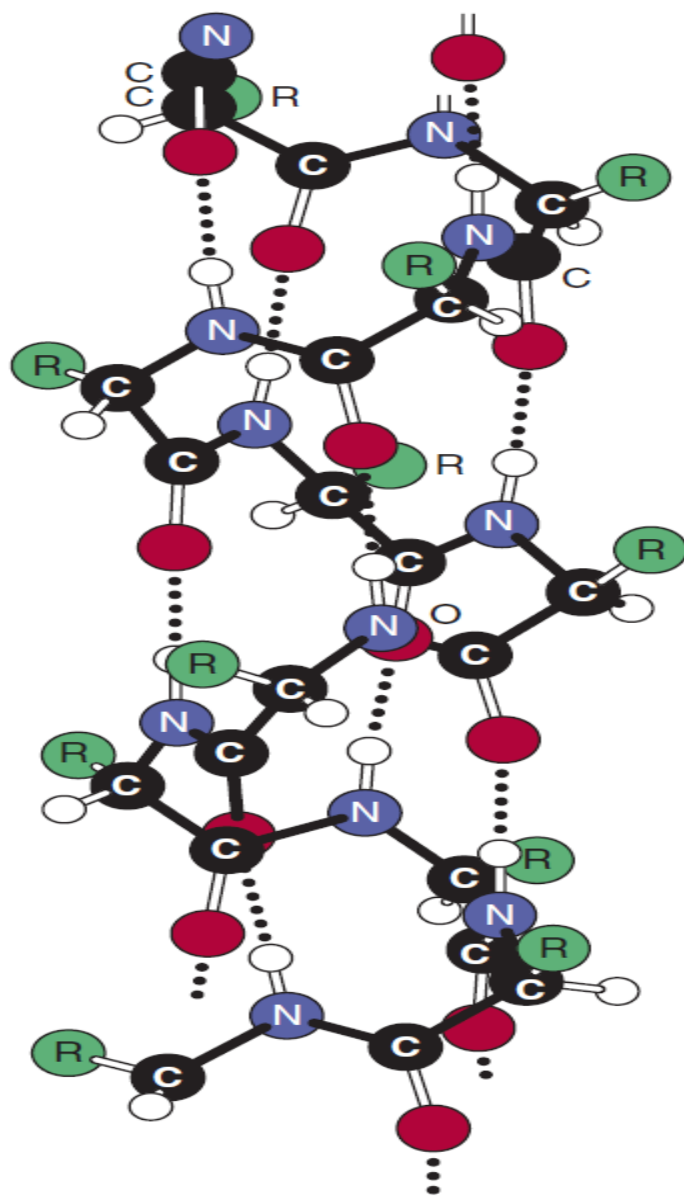




Alpha Helix

- ▶ The stability of an α helix arises primarily from hydrogen bonds formed between the oxygen of the peptide bond carbonyl and the hydrogen atom of the peptide bond nitrogen of the fourth residue down the polypeptide chain.





constraints affect the stability of an α *helix*

- (1) *the intrinsic propensity of an amino acid residue to form an α helix;*
- (2) *the interactions between R groups, particularly those spaced three (or four) residues apart;*
- (3) *the bulkiness of adjacent R groups;*
- (4) *the occurrence of Pro and Gly residues;*
- (5) *interactions between amino acid residues at the ends of the helical segment and the electric dipole inherent to the helix.*



Beta Sheet

- ▶ The amino acid residues of a β sheet, when viewed edge-on, form a zigzag or pleated pattern in which the R groups of adjacent residues project in opposite directions.



Beta Sheet

- ▶ derive much of their stability from hydrogen bonds between the carbonyl oxygens and amide hydrogens of peptide bonds.
- ▶ Schematic diagrams represent β sheets as arrows that point in the amino to the carboxyl terminal direction.

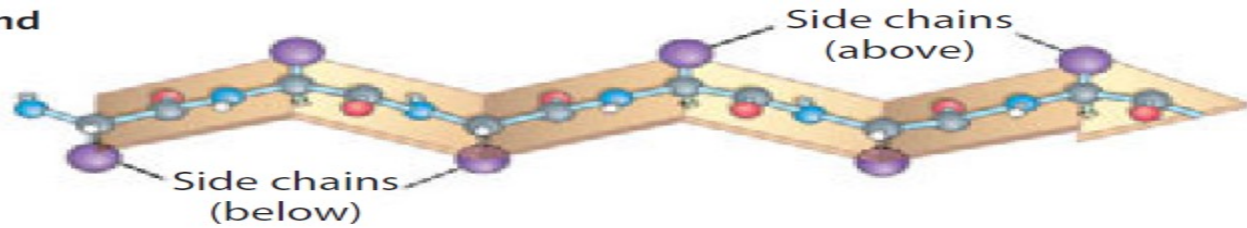




β pleated sheet

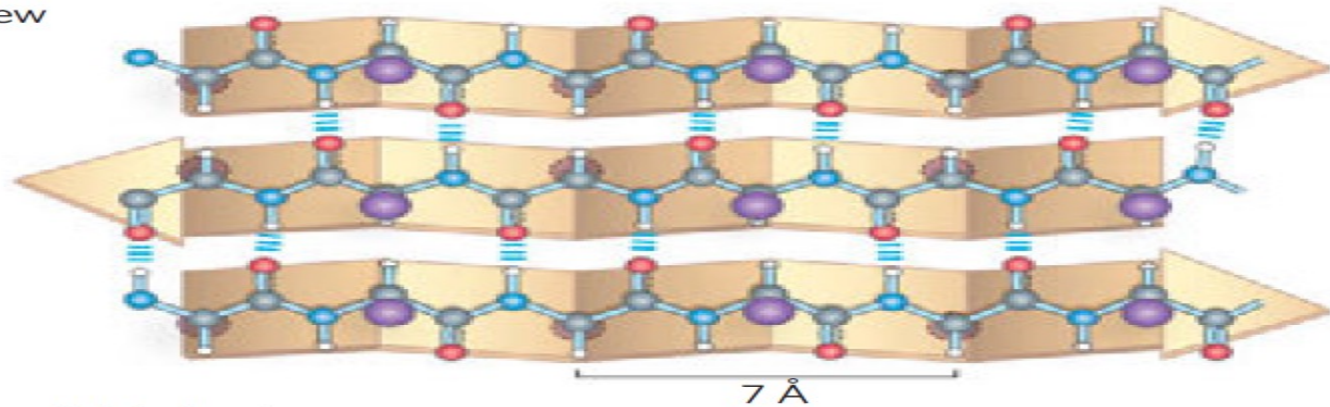
(a) β strand

Side view



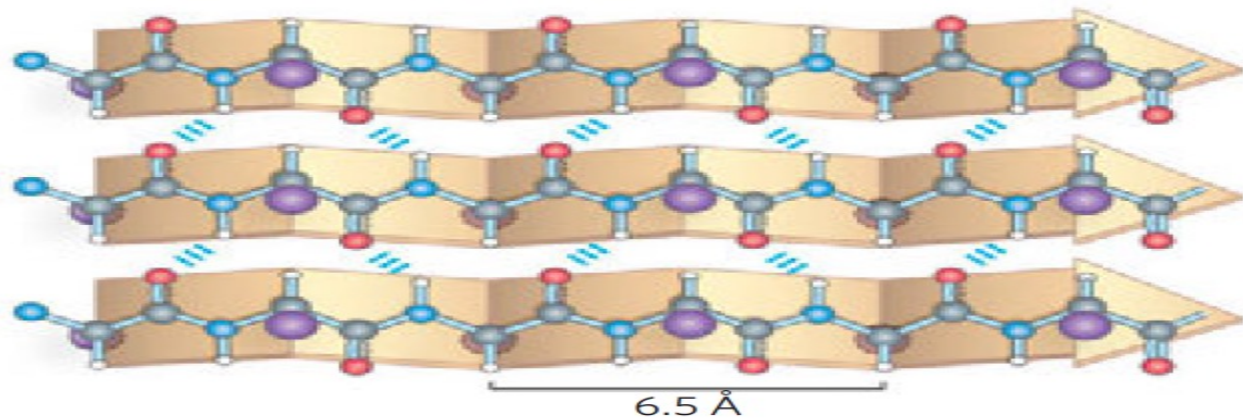
(b) Antiparallel β sheet

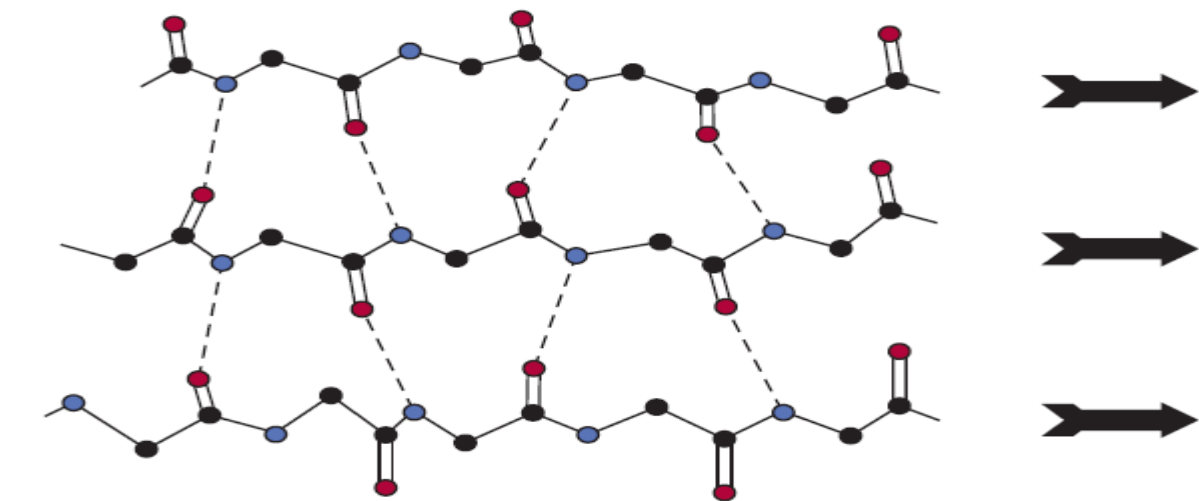
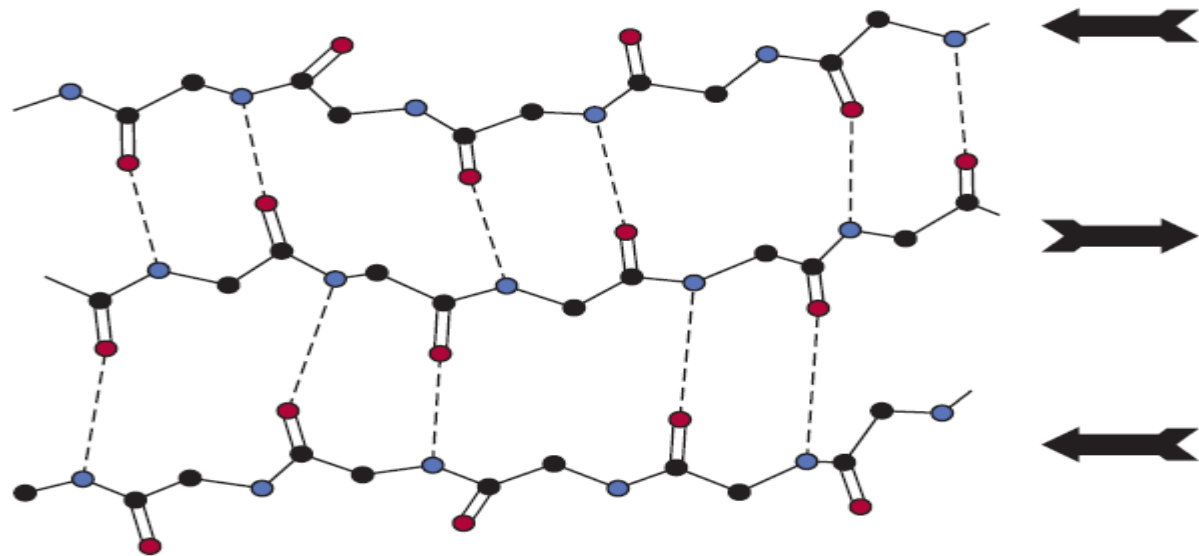
Top view



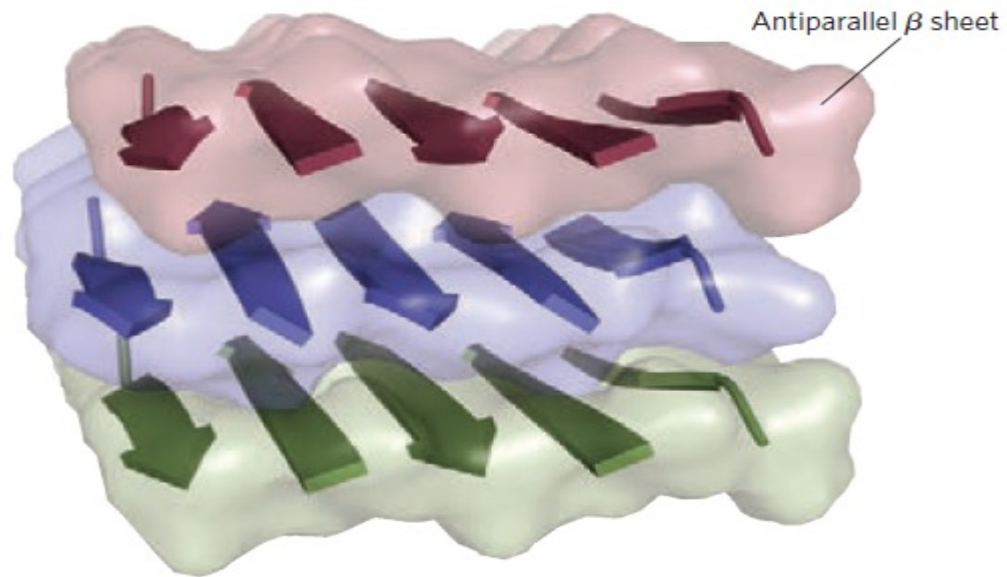
(c) Parallel β sheet

Top view





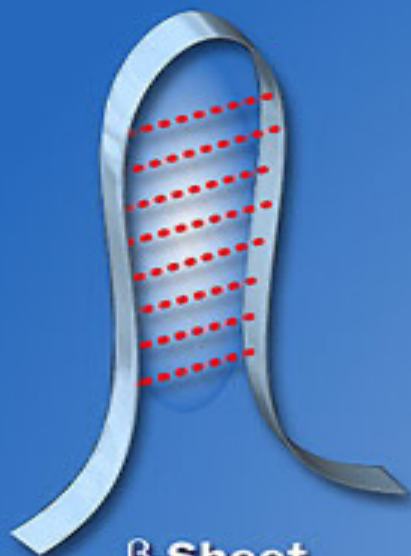
Silk



Hydrogen bonding



α -Helix



β -Sheet



Electrostatic attraction



Hydrophobic interaction



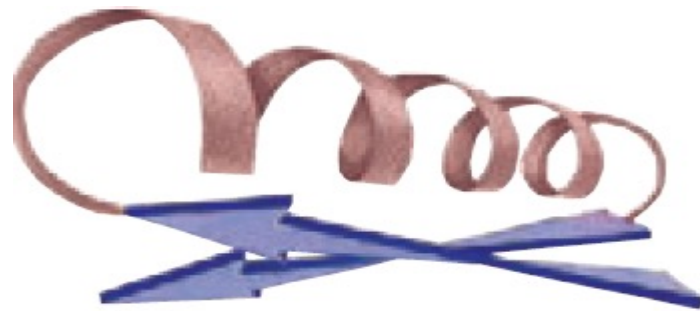
Metal-ion coordination

-
- ▶ Turns and bends refer to short segments of amino acids that join two units of the secondary structure
 - ▶ Loops are regions that contain residues beyond the minimum number necessary to connect adjacent regions of secondary structure





β Barrel



β - α - β Loop

Tertiary structure

- ▶ refers to the entire three-dimensional conformation of a polypeptide



Quaternary structure

- ▶ defines the polypeptide composition of a protein and, for an oligomeric protein, the spatial relationships between its protomers or subunits



-
- ▶ Oligomeric proteins can be composed of multiple identical polypeptide chains or multiple distinct polypeptide chains.
 - ▶ Proteins with identical subunits are termed homooligomers
 - ▶ Proteins containing several distinct polypeptide chains are termed heterooligomers



-
- ▶ Higher orders of protein structure are stabilized primarily—and often exclusively—by noncovalent interactions.
 - ▶ hydrophobic interactions, hydrogen bonds and salt bridges, Electrostatic Forces, van der Waals Forces, some covalent disulfide bonds



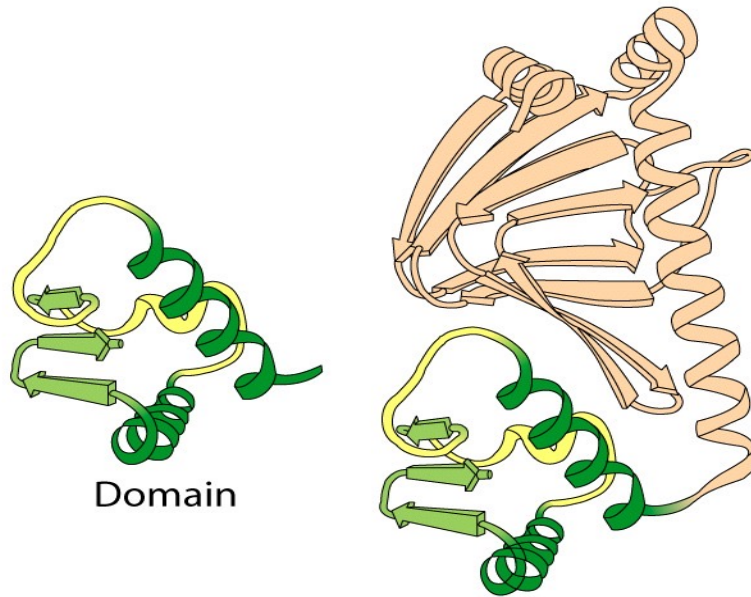
– Lys – Ala – His – Gly – Lys – Lys – Val – Leu

Amino acid sequence
of polypeptide chain

PRIMARY STRUCTURE



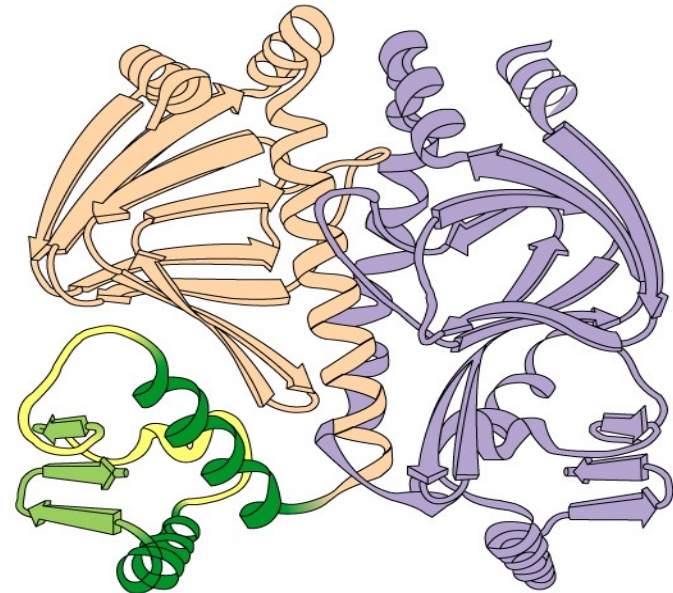
SECONDARY STRUCTURE



Domain

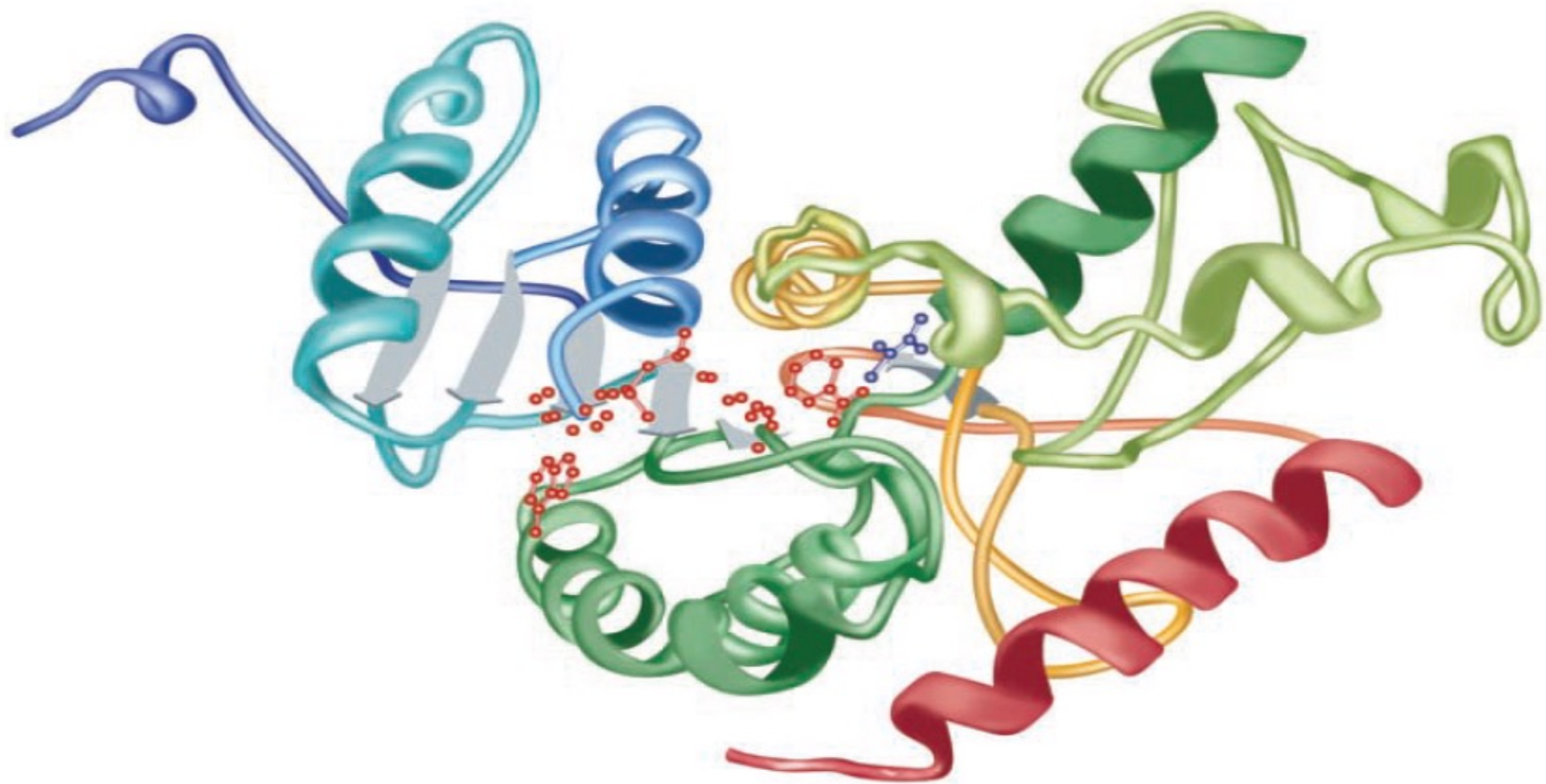
Monomeric polypeptide molecule

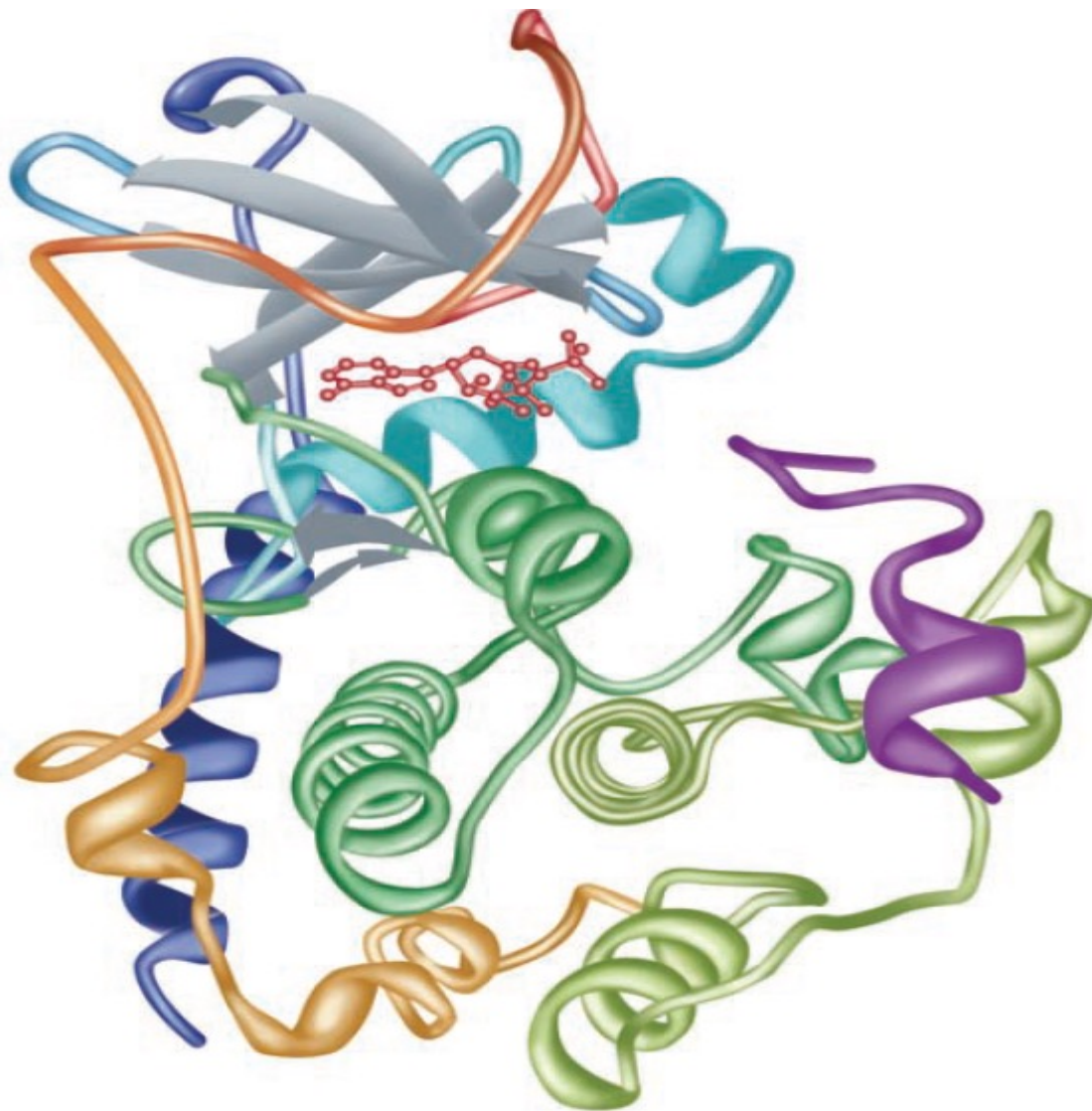
TERTIARY STRUCTURE



Dimeric protein molecule

QUATERNARY STRUCTURE



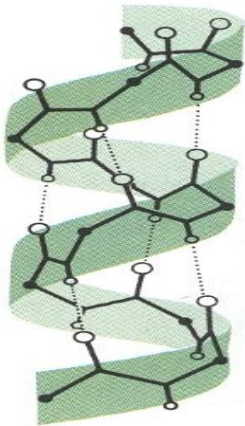


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- In general proteins fold into two broad classes of structure termed, globular proteins or fibrous proteins
 - Globular proteins are compactly folded and coiled, whereas, fibrous proteins are more filamentous or elongated

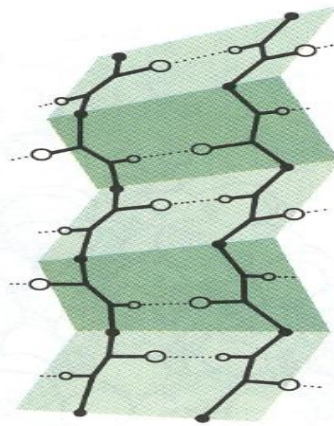




(a) Primary structure (amino acid sequence in the protein chain)

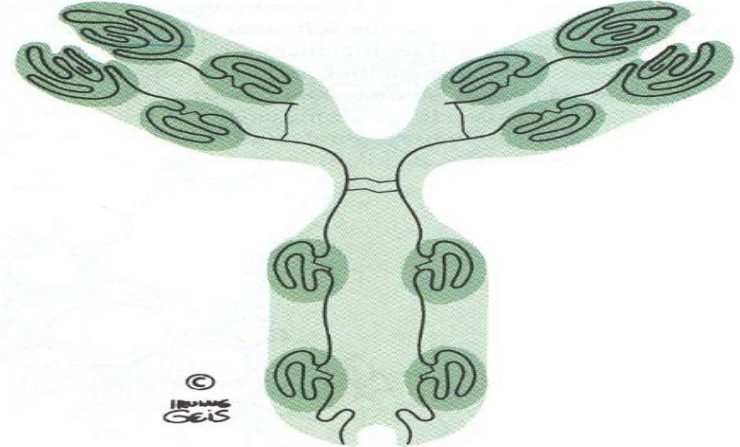


α helix



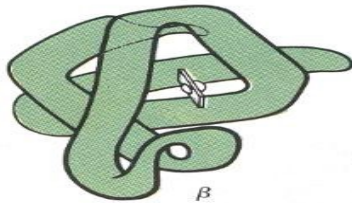
β sheet

(b) Secondary structure



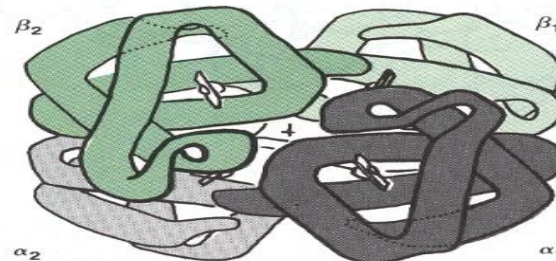
Domains (dark color) in an antibody molecule

(c) Local folding



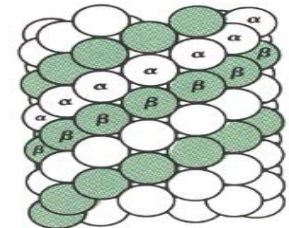
One complete protein chain (β chain of hemoglobin)

(d) Tertiary structure



The four separate chains of hemoglobin assembled into an oligomeric protein

(e) Quaternary structure



α (white) and β (color) tubulin molecules in a microtubule

(f) Quaternary structure

Denaturation of proteins

- ▶ Loss of all structures of protein except the primary structure.
- ▶ Break of all bonds except the peptide and disulfide bonds.
- ▶ Loss of protein function.
- ▶ 1.Acids 2.Alkalis 3.Metals 4.Temperature



Hemoglobin and myoglobin

- ▶ The study of myoglobin and hemoglobin demonstrates the close relationship between the structure and function of proteins.
- ▶ Myoglobin and hemoglobin are members of heme containing proteins.



Myoglobin

- ▶ heme containing protein found in red muscle and functions as temporal oxygen storage molecule.
- ▶ The primary structure is a single polypeptide chain containing 153 amino acid residues



Myoglobin

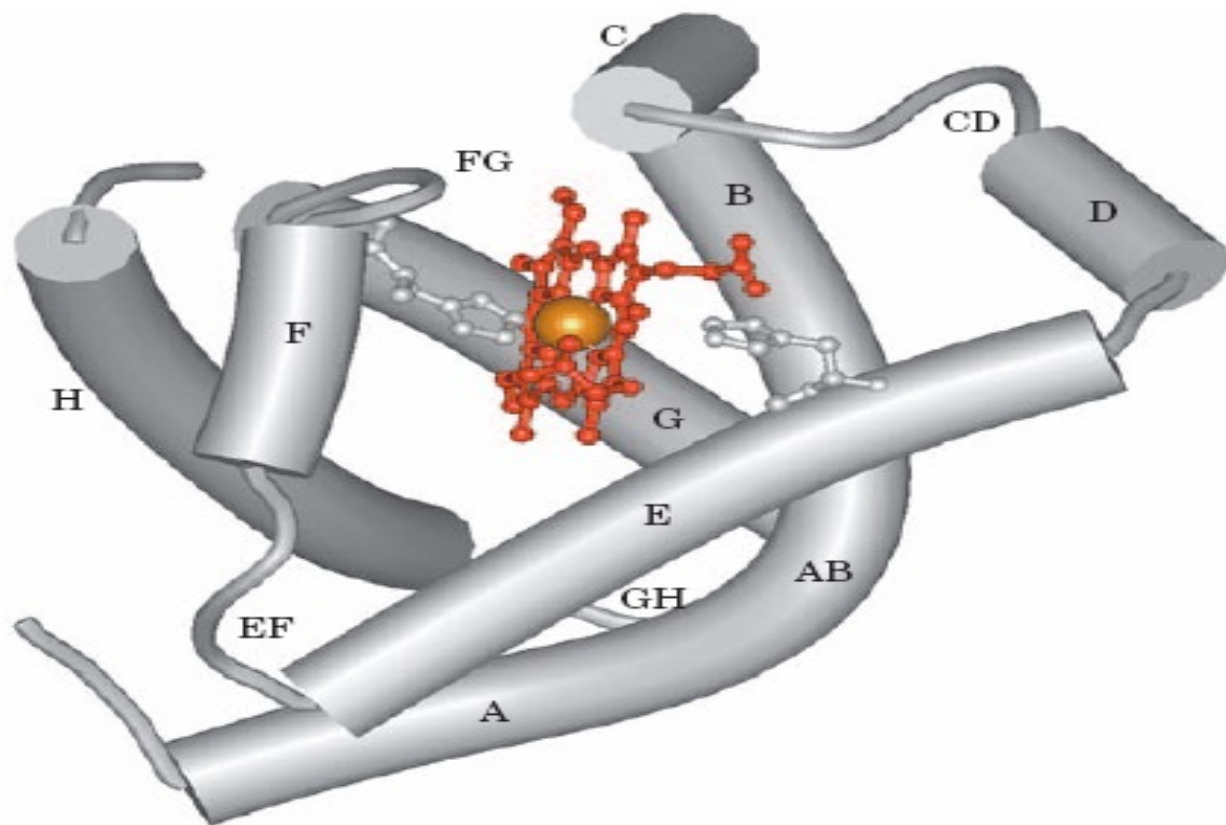
- ▶ The secondary structure is formed exclusively from alpha helices(75%).
- ▶ There are eight alpha helix units
- ▶ The helices are named from A to H.

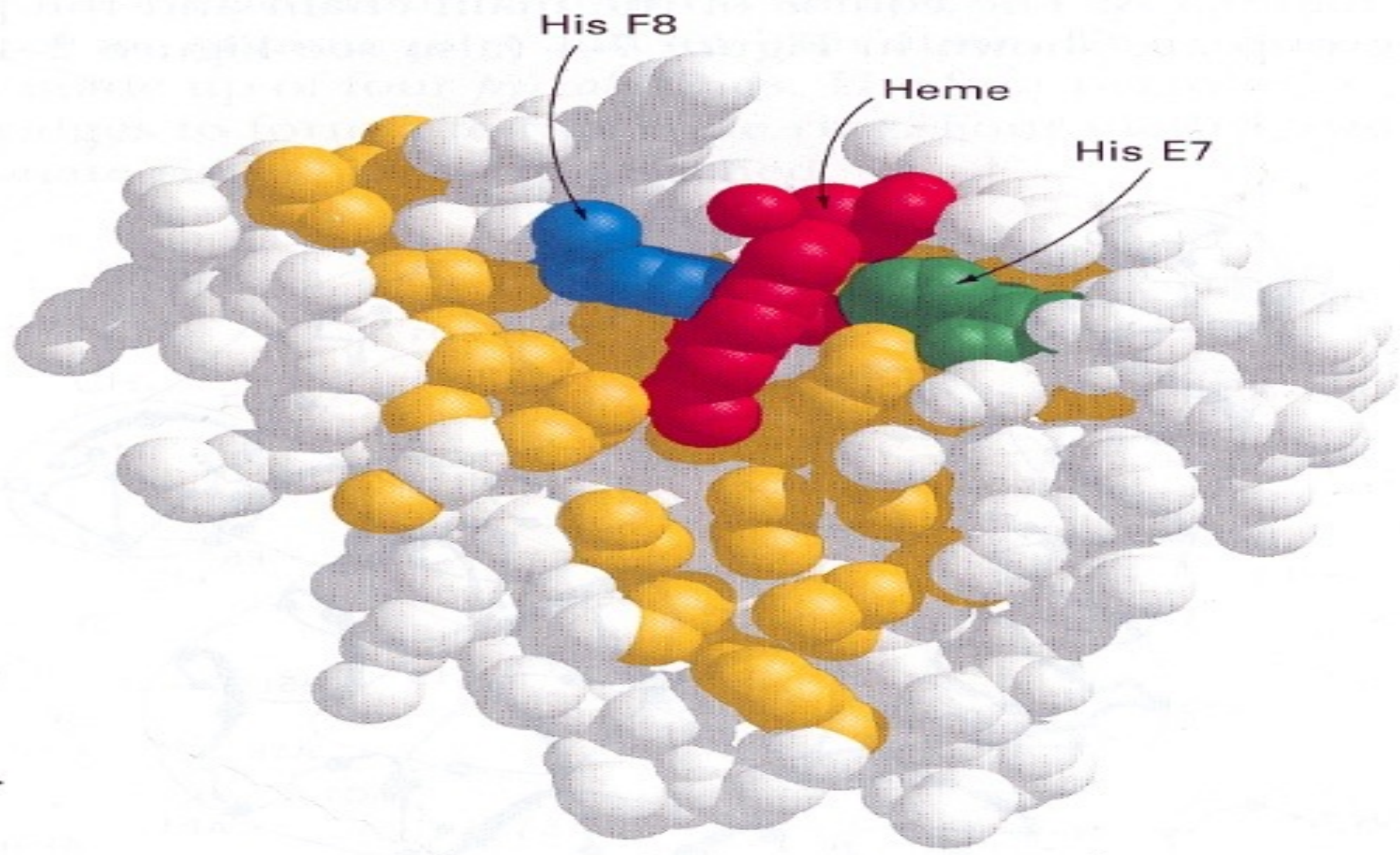


Myoglobin

- ▶ The protein is globular nearly spherical in outline. The polar amino acids are located on the exterior while nonpolar residues are located in the interior except for two histidine residues near the heme group (hist F8 and hist E7) which serve an important role in oxygen binding.



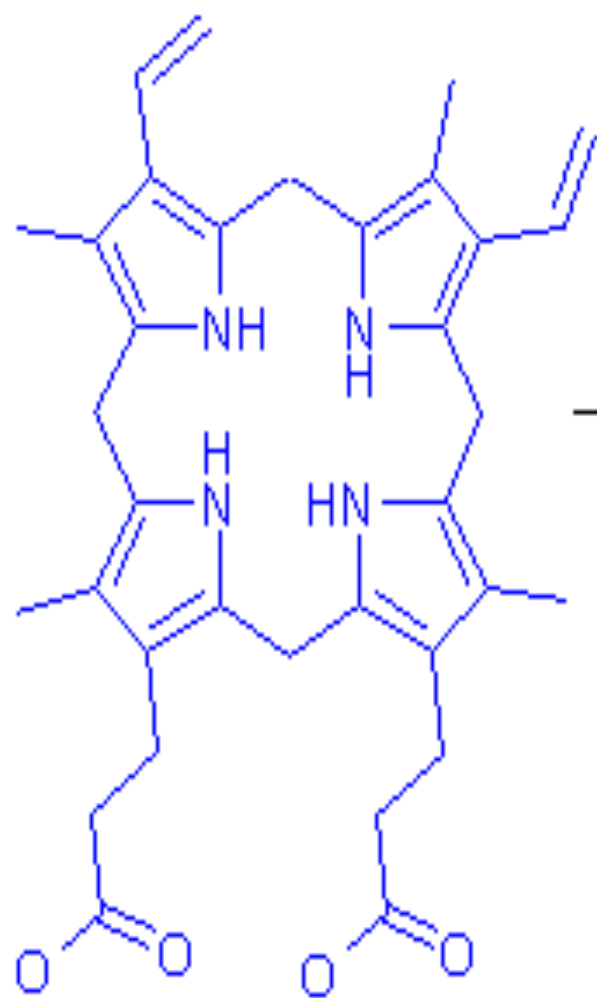




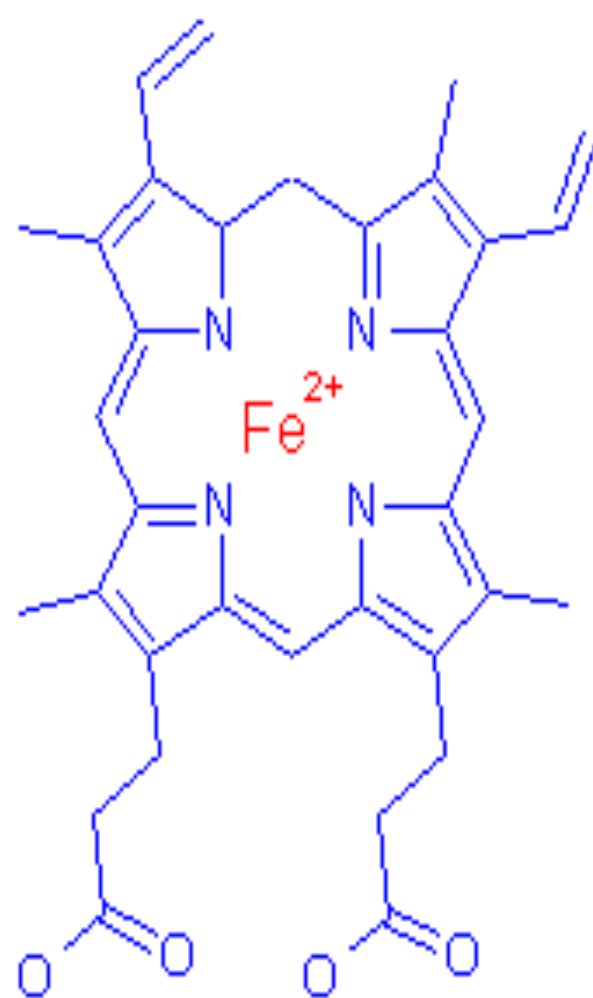
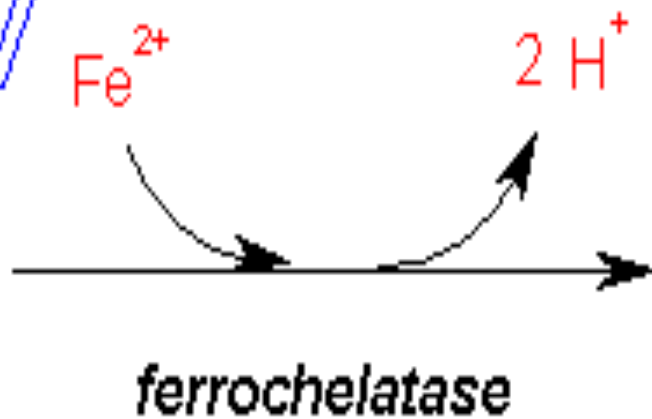
HEME

- ▶ A tetrapyrrole ring or a complex of protoporphyrin IX with central ferrous atom that is attached to the four nitrogen atoms of the porphyrin ring.



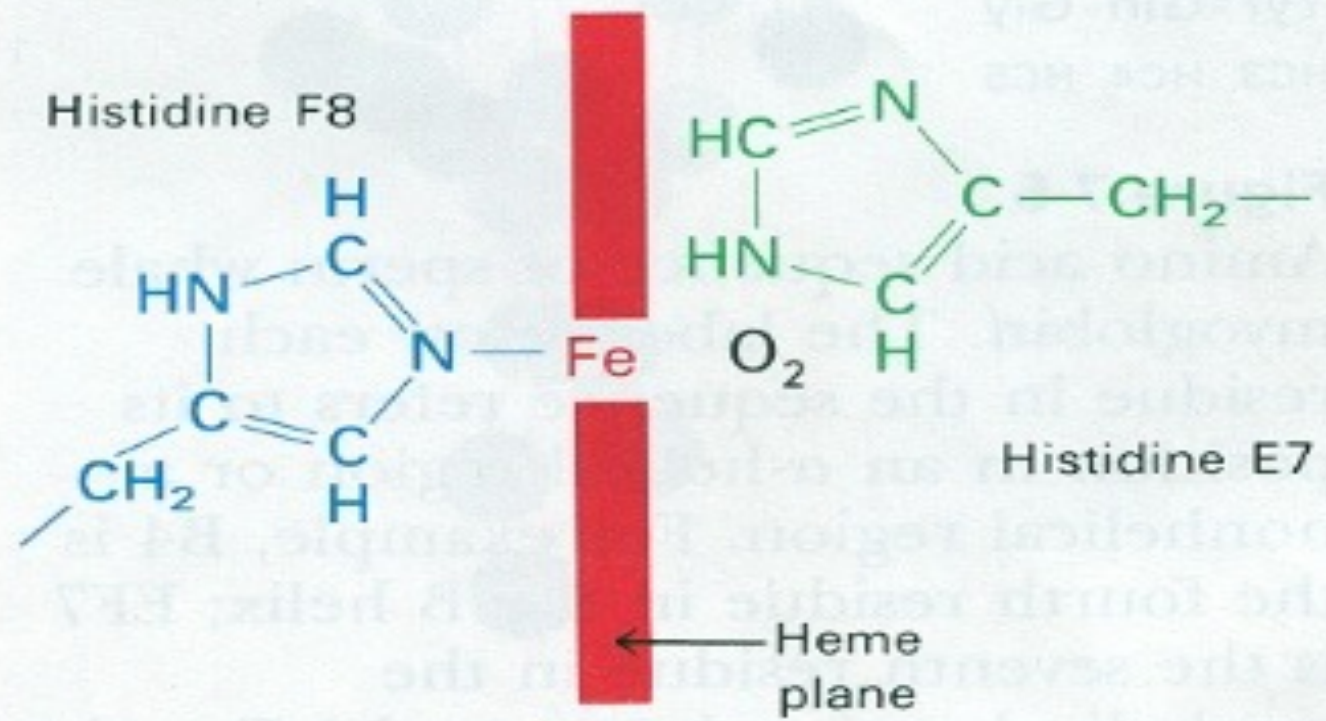


protoporphyrin IX

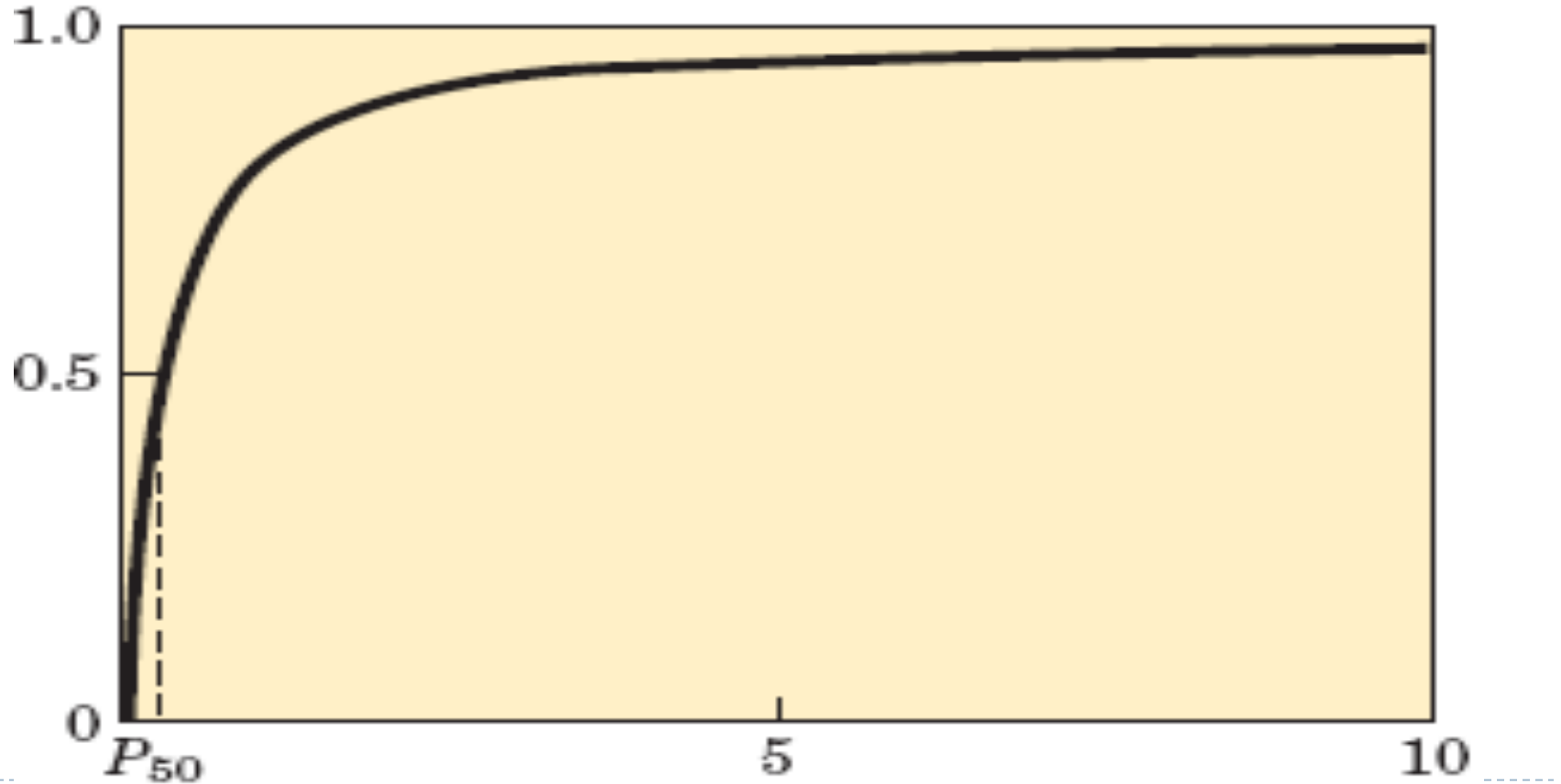


heme

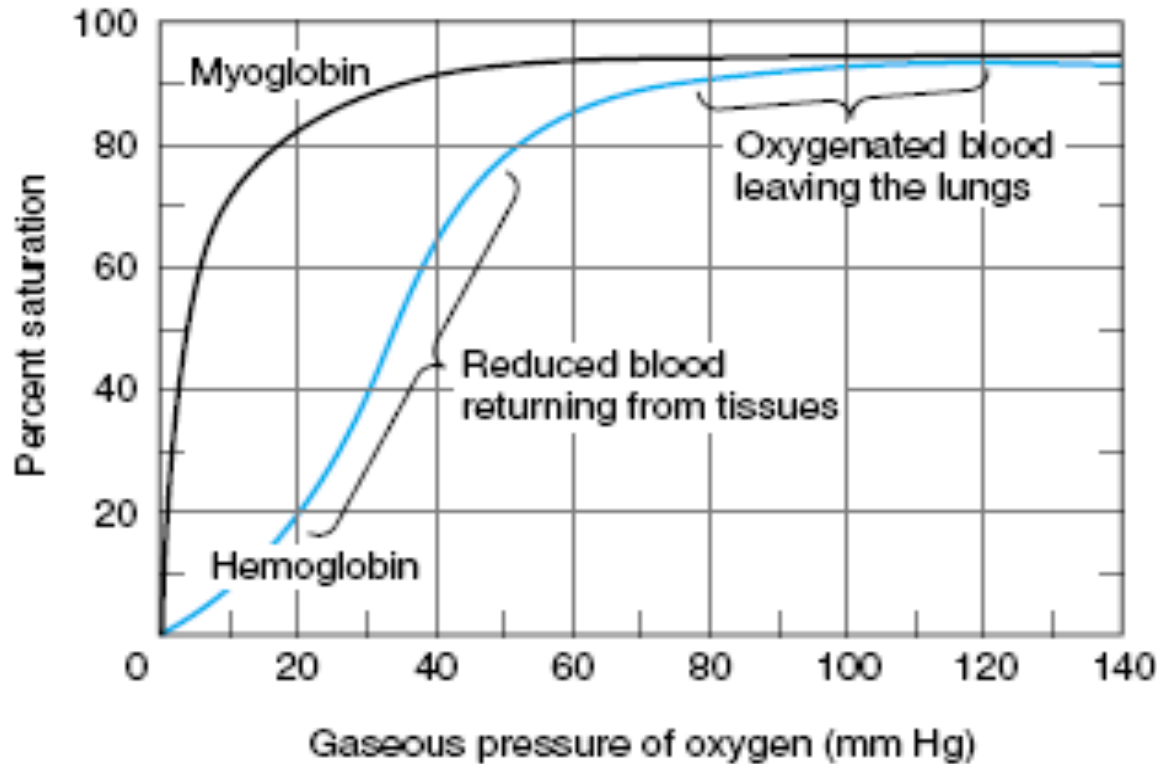
► The function of myoglobin is to store oxygen in muscle for usage during



- ▶ **Heme in myoglobin has higher affinity for oxygen than hemoglobin.**



Heme in myoglobin has higher affinity for oxygen than hemoglobin.



Hemoglobin

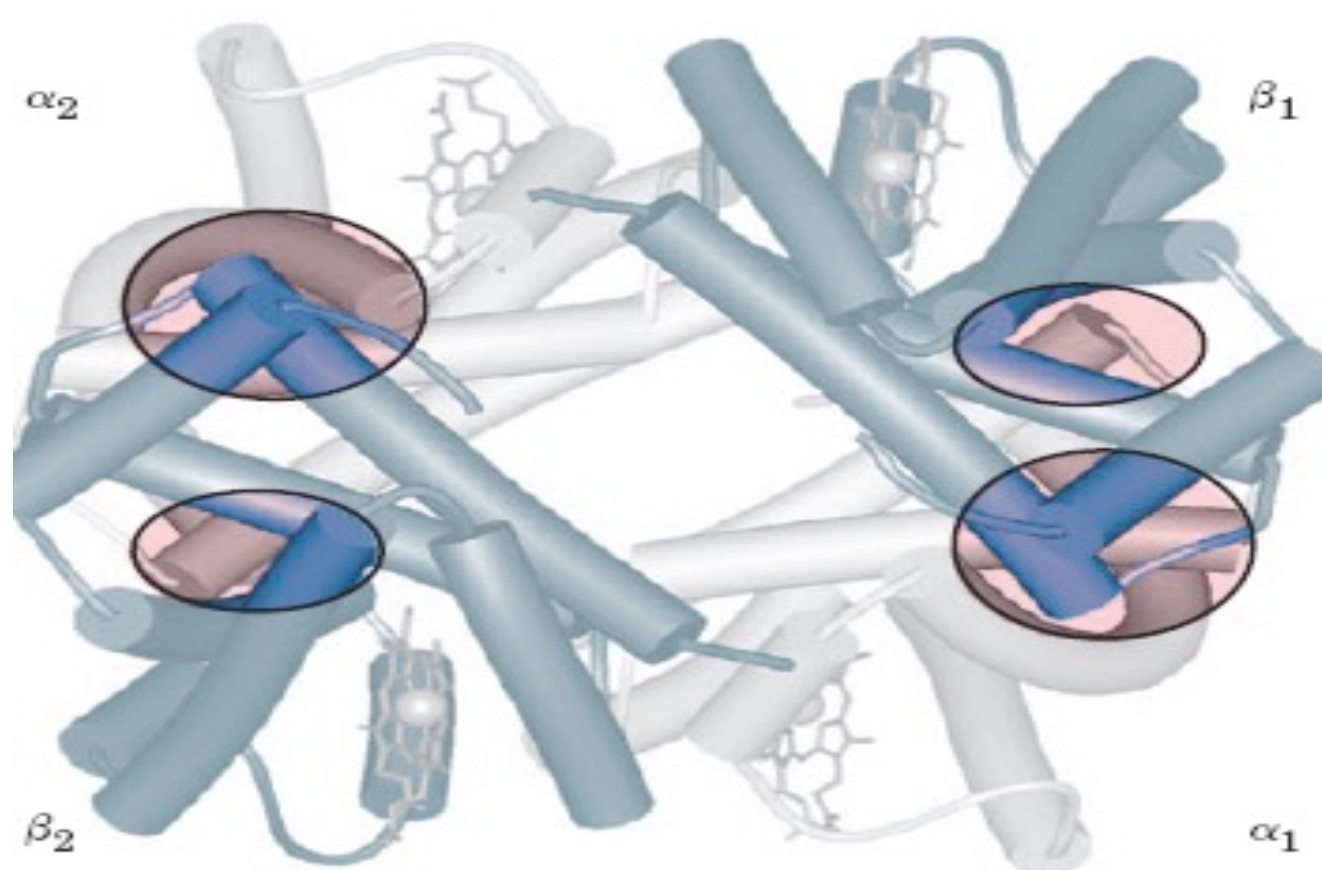
- ▶ is a tetrameric heme containing protein
- ▶ In adult contain 2 alpha and 2 beta chain
- ▶ Alpha chain has 141 amino acids residue folded into seven helical regions.
- ▶ Beta chain has 146 amino acid residue and has identical secondary and tertiary structure to myoglobin with eight helical regions
- ▶ Each chain contains one heme molecule in its cavity
- ▶ The four polypeptides are connected by noncovalent bonds.



Hemoglobin

- ▶ Types of hemoglobin:
- ▶ Fetal Hb: Alpha 2 Gamma 2
- ▶ HbA1: Alpha 2 Beta 2
- ▶ Hb A2: Alpha2 Delta 2

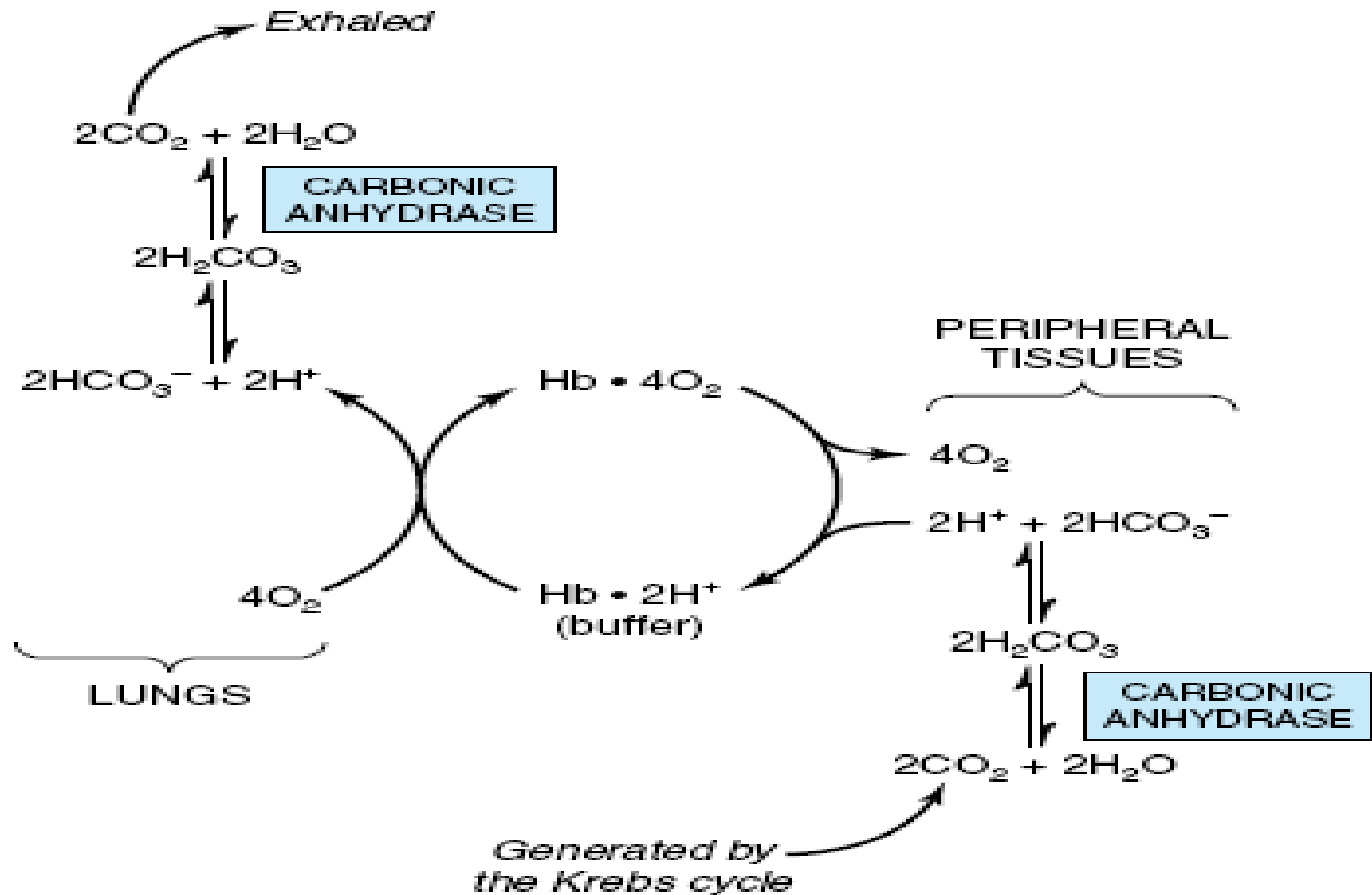




Hemoglobin

- ▶ Hb transports oxygen from the lung to the tissue
- ▶ Hemoglobin transport CO_2 and H^+ from tissues to the lung.
- ▶ For each two O_2 molecules released hemoglobin will carry one proton contributing to the buffering capacity of the blood
- ▶ The lower pH in the tissues will favor the oxygen delivery to the tissues whereas in the lung the reverse is true; this reversible process is called Bohr effect
- ▶ Bohr effect depend on the cooperative function of hemoglobin





Hemoglobin Oxygen Dissociation Curve

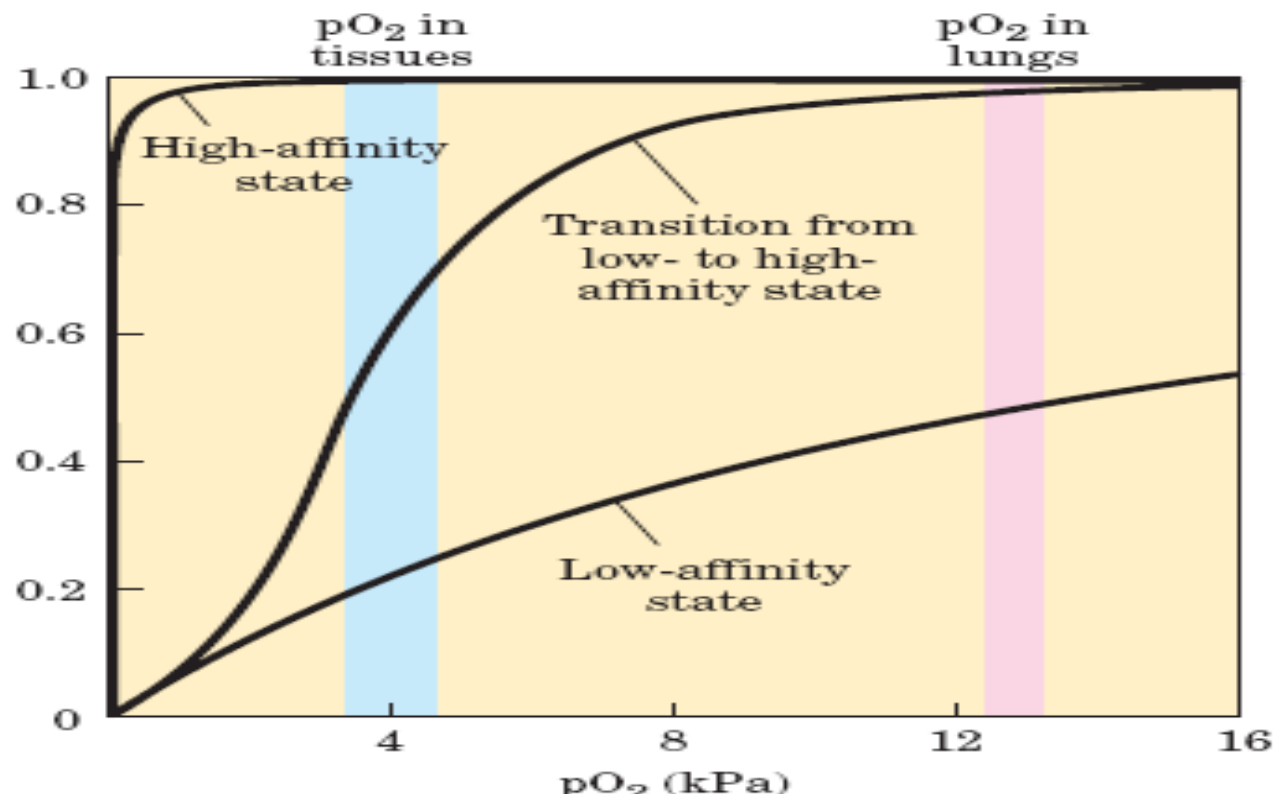
- ▶ sigmoid in shape compared to the hyperbolic curve of myoglobin.
- ▶ This will allow hemoglobin to serve as an efficient transporter of oxygen because there is marked change in oxygen saturation of hemoglobin in response to change in partial oxygen pressure.



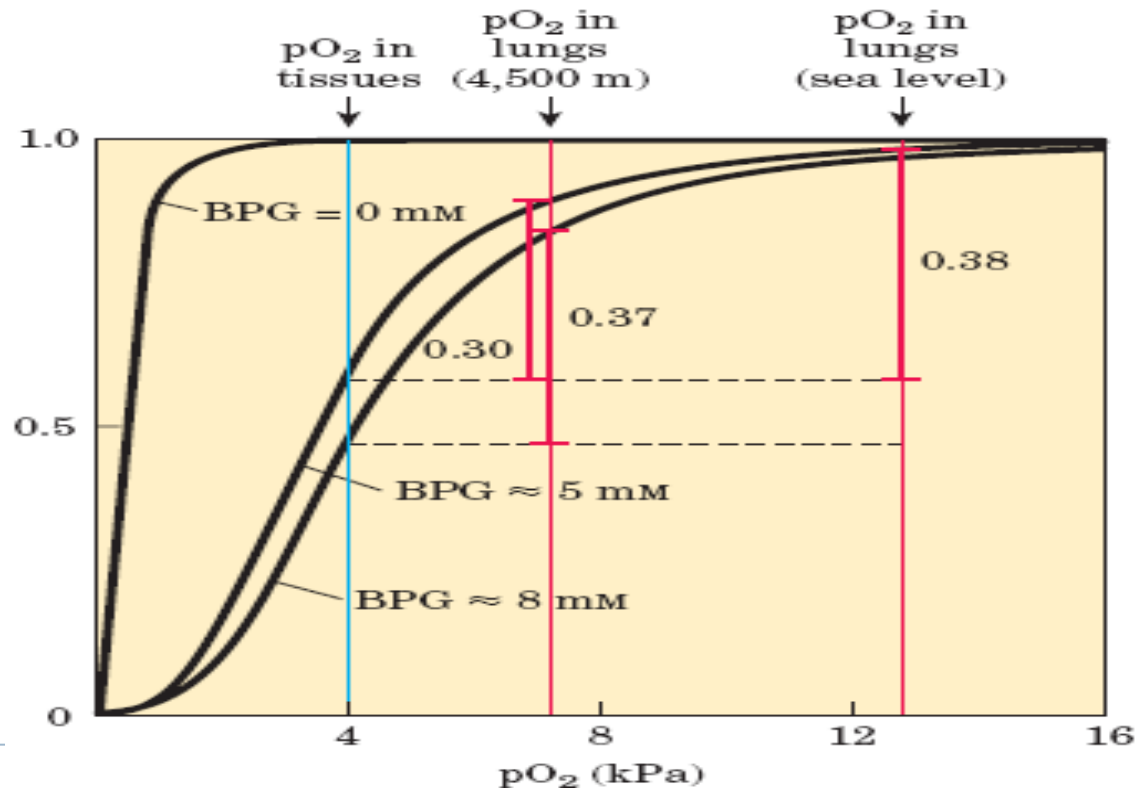
Hemoglobin Oxygen Dissociation Curve

- ▶ P50 is the partial oxygen pressure that half saturates a given hemoglobin
- ▶ P50 for HbA is 26 mm Hg
- ▶ P50 for HbF is 20 mm Hg





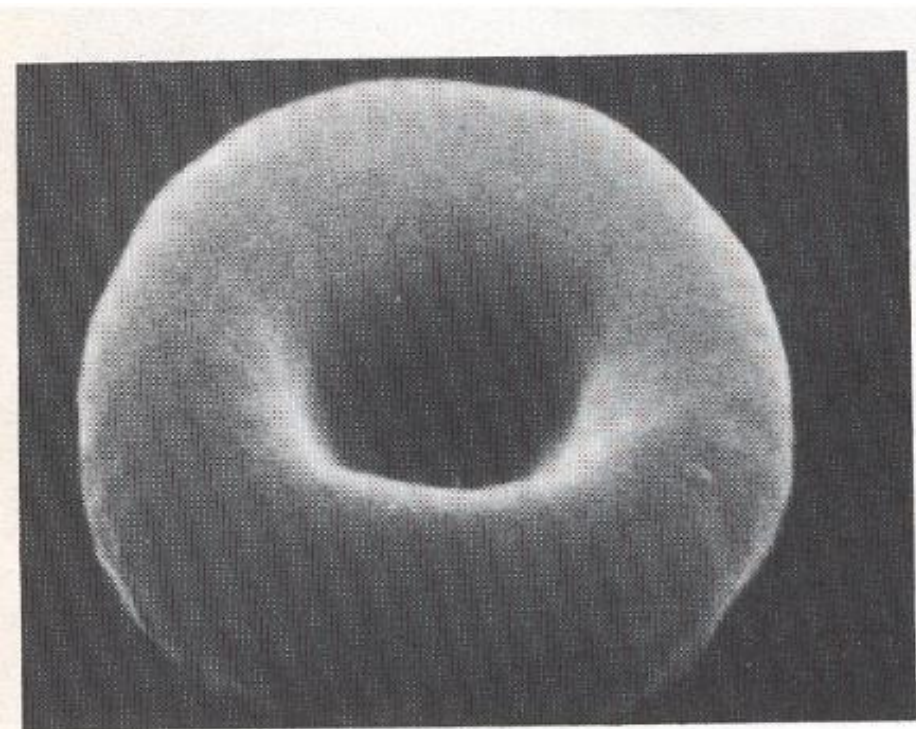
Stabilizes the Structure of Hemoglobin



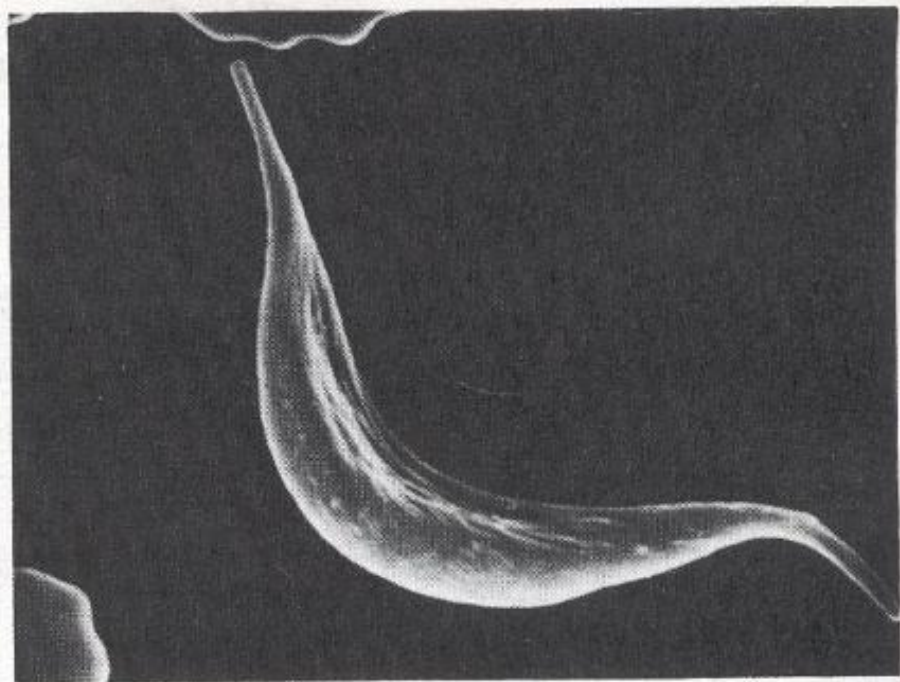
Hemoglobinopathies

- ▶ Sickle Cell Hemoglobin
- ▶ Due to single amino acid change in the Beta chain in position 6 valine replaces glutamate
- ▶ Deoxyhemoglobin S is less soluble and precipitate forming long fiber leading to deformity of the red cells and hemolytic anemia
- ▶ Diagnosed by electrophoresis





A



B



Hemoglobinopathies

▶ Thalassemia

Group of diseases characterized by partial or complete absence of one of the polypeptide either alpha or beta chain due to deletion mutations



Collagen

- ▶ Is a Fibrous Protein
- ▶ Collagen is the most abundant of the fibrous proteins that constitute more than 25% of the protein mass in the human body.
- ▶ These proteins represent a primary source of structural strength for cells and tissues.



Amino acid
sequence

– Gly – X – Y – Gly – X – Y – Gly – X – Y –

2° structure



Triple helix



Collagen

- ▶ is initially synthesized as a larger precursor polypeptide, procollagen. Numerous proline and lysine residues of procollagen are hydroxylated by enzymes that require ascorbic acid (vitamin C). Hydroxyproline and hydroxylysine residues provide additional hydrogen bonding capability that stabilizes the mature protein.
- ▶ The best-known defect in collagen biosynthesis is scurvy, a result of a dietary deficiency of vitamin C.

